

Breaking Bad Opioid Sorting*

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Abstract

Consumers frequently rely on experts to access goods and services. In these markets, consumer and expert preferences over outcomes are linked through consumers' choice of experts, or sorting. We investigate how consumer sorting contributes to observed outcome differences and its implications for expert-targeted policies. Using rich employer-sponsored health insurance claims data on opioid prescriptions for chronic pain, we first show that prescription intensity is highly dispersed across physicians. Decomposing the variance of physicians' prescription decisions, we find patient sorting three times as important as physicians' inherent propensity for prescription intensity dispersion, and almost exclusively driven by factors not concordant with CDC clinical guidelines. We develop and estimate an equilibrium model of patients' physician choice and physicians' prescription decisions, in which patients optimally choose physicians based on both their opioid preferences and their expectations of physicians' prescription decisions. Our counterfactual analysis shows that expert-targeted policies to curb clinically unwarranted opioid prescribing are severely attenuated by patient resorting. We propose an alternative expert-targeted policy that eliminates sorting based on preferences exceeding medical consensus or CDC guideline, while preserving appropriate care for patients with medical needs for prescription opioids.

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1 Introduction

In many markets, consumers rely on experts for access to goods and services. For instance, investors turn to asset managers to access complex investment opportunities, and patients consult physicians for medical advice and treatment. The stakes of choosing the right expert can be substantial: selecting the wrong asset manager may expose investors to levels of financial risk that greatly differ from their preferences, and choosing the wrong physician can significantly impact long-term health. Given the high stakes, consumers have strong incentives to sort into experts, choosing an expert whose preferences, style, or philosophy closely align with their own. As a result, market outcomes reflect both expert preferences and consumer preferences, linked together through consumer sorting. Despite the importance of disentangling these forces for policy design, empirical evidence quantifying the contribution of each channel to observed outcomes remains extremely limited.

In this paper, we study the importance of consumer sorting versus expert preferences for observed outcome differences and their implications for expert-targeted policies. We focus on opioid prescribing for chronic pain patients in the United States. Chronic pain affects 60 million adults in the U.S. (Lucas and Sohi, 2024). One-fifth of U.S. chronic pain patients rely on prescription opioids through their chosen physicians (Dahlhamer et al., 2021). The setting is important for two reasons. First, contrary to acute pain, chronic pain involves continuous episodes of pain management, exposing patients to substantially higher risk of opioid misuse. Patient sorting into high-prescribing physicians may significantly exacerbate these risks. Second, despite the prevalence of chronic pain, the role of patient sorting versus physician preference for opioid prescribing to chronic pain patients remains widely understudied. Prior work has focused almost exclusively on quasi-random assignment of patients to physicians, such as acute pain patients admitted through the Emergency Department, to identify only the role of physicians for opioid prescribing.

Using employer-sponsored health insurance claims data, we show that patients’ choice of physicians, or patient sorting, is three times as important in explaining the observed dispersion in opioid prescribing rates across physicians as physicians’ inherent prescribing propensity. Moreover, patient sorting is almost exclusively driven by factors not concordant with CDC clinical guidelines for prescribing opioids (Dowell et al., 2016, henceforth CDC2016). This finding suggests that policies targeting top-prescribing physicians, such as the Drug Enforcement Administration’s prescription license revocation policy, aimed at reducing clinically unwarranted opioid prescribing, may be severely attenuated by patient re-sorting. To understand how patient sorting interacts with top physician-targeted policies, we develop and estimate an equilibrium model of patient sorting and physician prescribing. We

use our estimates to simulate several counterfactuals. First, mimicking top expert-targeted policies, we consider a counterfactual in which we exclude the top 10% of physicians by risk-adjusted prescription rate from the market. Despite being extreme, this policy reduces aggregate prescribing by only 11%. We show that this result is driven by affected patients resorting to non-excluded physicians with similar prescription propensities. In other words, resorting severely attenuates the exclusion policy. Second, eliminating resorting, we consider the same exclusion policy but allow affected patients to only choose between random assignment to a physician or to forgo care. Without resorting, aggregate prescribing would fall by an additional 11 percentage points. However, randomization breaks sorting indiscriminately, regardless of patients’ medical needs. To address this issue, we consider a counterfactual in which we force physicians to disregard any patient opioid preference exceeding medical consensus or CDC guidelines (CDC2016). This eliminates patients’ incentives to sort based on preferences exceeding medical consensus or CDC guidelines. Breaking only bad sorting in our exclusion policy, aggregate opioid prescriptions would drop by 13%–21%, depending on how much medical consensus was considered appropriate (e.g., compassionate prescribing for pain), while preserving treatment for patients with medical needs.

We establish these findings using health insurance claims data from the Merative MarketScan Research Databases for the New York City Metropolitan Statistical Area over the period 2016–2022. These visit-level data allow us to link patients, their chosen physicians, and any resulting opioid prescriptions. We begin by documenting that physicians’ prescription rates are highly dispersed, even after controlling for a rich set of patient severity measures. The top 10% of physicians in terms of prescribing intensity account for approximately 30% of opioid prescriptions. This pattern aligns with the well-documented heterogeneity in physicians’ prescription intensity reported in the literature (e.g., Finkelstein et al., 2016).

Next, we assess the importance of patient sorting for the dispersion of physicians’ prescription rates. Following the well-known two-way fixed effects approach developed by Abowd et al. (1999), we decompose the variance of physicians’ prescription decisions into components attributable to patient fixed effects, physician fixed effects, and patient sorting. There are two identification challenges. First, physician fixed effects are identified from patients who switch physicians. When switching is limited, fixed effects estimates can be severely biased, a problem known as limited mobility bias (Andrews et al., 2012). We follow recent advances in the labor literature to address limited mobility bias by grouping physicians and patients using k-means clustering and estimating group-level fixed effects (Bonhomme et al., 2019; Mourot, 2025). Second, fixed effects estimates may be biased if patients’ switching decisions are due to time-varying characteristics unobserved by the econometrician. We provide evidence to mitigate this concern following Card et al. (2018).

Using the two-way fixed effects estimator, we find that patient sorting is three times as important as physicians’ inherent prescribing propensity in explaining the observed dispersion in physicians’ opioid prescribing rates. Patient sorting is almost exclusively driven by patient preferences not concordant with the CDC guideline for prescribing opioids for chronic pain (CDC2016). Our findings are consistent with anecdotal evidence from online forums, where patients with strong preferences for opioids actively seek and share information about physicians’ propensity to prescribe.¹ Moreover, our findings suggest that policies that target only a subset of physicians may be severely attenuated when ignoring the effect of patient resorting.

To understand how patient sorting interacts with top physician-targeted policies, we develop an equilibrium model of patient choice of physicians and physicians’ opioid prescription decisions. Patients’ choice stage is a static discrete choice demand model in the spirit of Berry et al. (1995). We model patient types as time-persistent tuples that summarize both opioid preferences and information. Patients’ information types determine how accurately patients perceive physicians’ prescription propensities. When choosing a physician, patients consider the expected prescription probability based on their own opioid preferences and their perceptions about each physician’s prescribing propensity. In the prescription stage, partially altruistic physicians optimally prescribe opioids based on their own prescribing propensities and patients’ opioid preferences.

Estimating our model is challenging because it features three dimensions of unobserved heterogeneity: physicians’ prescription propensity, patients’ opioid preferences, and patients’ information types. To reduce the estimation burden, we estimate our model in three steps using a limited-information maximum-likelihood approach. First, we use our panel on realized opioid prescriptions to estimate both physicians’ prescription propensities and patients’ opioid preferences with a two-way fixed effects logit model. This estimation closely mirrors our earlier decomposition of prescription rate variation and follows the grouped fixed effects approach (Bonhomme et al., 2019; Mourot, 2025). This approach addresses both limited mobility and incidental parameter biases. Second, conditional on the first-stage estimates, we estimate the patient choice stage by constrained maximum likelihood, imposing a share constraint commonly relied on in discrete choice demand models (Berry et al., 1995). This step recovers patients’ mean utilities for each physician, parameters governing patient preferences for office visits, and the variance of time-invariant patient information types by leveraging the panel structure of the data. Finally, we decompose the estimates for patient

¹For example, https://www.reddit.com/r/hudsonvalley/comments/1ccgimp/pain_management_doctor_that_will_actually_manage/ (accessed 05/07/2026), and https://www.reddit.com/r/grassvalley/comments/1kn7h0i/doctor_that_prescribes_opiates_in_town_or_nc/ (accessed 05/07/2026)

opioid preferences into components that are concordant with the CDC guideline, components that warrant prescribing based on medical consensus, such as patient comorbidities, and components that are neither in line with the CDC guideline nor medical consensus. To do so, we use a rich vector of patient-level controls, including patient severity measures and patients’ diagnosis histories.

Physician-targeted policies are an important policy instrument to curb excessive opioid prescribing. We use our estimates to simulate several counterfactuals to illustrate two points. First, we highlight the mechanism of how patient sorting interacts with top physician-targeted policies, such as the Drug Enforcement Administration (DEA)’s prescription license revocation policy. Second, we show how the effectiveness of physician-targeted policies can be improved by accounting for patients’ equilibrium resorting in response to these policies. To do so, we consider an extreme version of top physician-targeted policies by excluding the top 10% of prescribers from the market. We find that even though our policy is extreme, aggregate opioid prescriptions drop by only 11%. This muted response arises because patients do not choose physicians randomly. Instead, our results imply that patients sort toward physicians whose prescribing propensity aligns with their own preferences for opioids. When top physician-targeted policies eliminate these first-best matches, patients optimally resort to the remaining physicians. Because high-prescribing physicians are disproportionately chosen by patients with strong preference for opioids, resorting disproportionately shifts demand toward the highest-prescribing remaining physicians, attenuating the aggregate reduction in opioid prescribing.

Next, we quantify the extent to which resorting attenuates our exclusion policy. In particular, we let patients who visited an excluded physician in the baseline choose only whether to seek care. Conditional on seeking care, patients are randomly assigned to a physician. Under this counterfactual, eliminating the top 10% of physicians by prescription rate reduces aggregate opioid prescriptions by 22%, an additional 11 percentage points drop compared to our previous counterfactual with resorting. These results demonstrate the importance of accounting for demand-side equilibrium responses in supply-side-focused policies targeting a subset of experts. Ignoring consumer choice attenuates the effectiveness of top expert-targeted policies.

Finally, we illustrate how top physician-targeted policies can be improved by accounting for patient equilibrium resorting. First, we note that restricting patient choice of physician via randomization has important drawbacks. Medical literature shows that the ability to choose a physician is crucial for patients to remain in care (Hsu et al., 2003). Consistent with these findings, our results predict that when patients are randomized to physicians, most affected patients would prefer to forgo care rather than to accept random assignment.

Although randomization aggressively reduces aggregate opioid prescriptions, it achieves so by breaking optimal patient-physician matches regardless of whether sorting is based on patient preferences concordant with the CDC guideline or medical consensus (“good sorting”), or not (“bad sorting”). This makes randomization an undesirable policy. Second, we show how top physician-targeted policies can leverage “good” sorting while breaking “bad” sorting, preserving patient choice and keeping patients in care. We consider a policy that requires physicians to prescribe only based on patient preferences concordant with CDC guidelines and medical consensus, but not others.² Since medical consensus is only partially observable, we bound the policy’s effect with two counterfactuals. First, we assume that all unobserved patient preferences do not support prescribing based on medical consensus. This shuts down all patient sorting based on unobserved preferences. By eliminating bad sorting, this policy reduces aggregate prescribing rates by 21%, a result close to randomization. This constitutes an upper bound on the effectiveness of our policy. Second, we repeat this exercise but now assume that all unobserved patient preferences support opioid prescribing based on medical consensus. In this case, aggregate prescriptions would drop by 13%. This is a lower bound on the effectiveness of our policy. Moreover, regardless of our assumptions on the unexplained part of patients’ opioid preferences, we show that the number of patient visits under this alternative policy closely matches the number of visits when we allowed for both good and bad sorting.

Our analysis contributes to a growing literature on the drivers of prescription opioid consumption and misuse in the U.S. Prior research has broadly focused on either the supply side or the demand side drivers. On the supply side, research has identified aggressive marketing of OxyContin as a key driver of the opioid crisis (Alpert et al., 2021), highlighted the role of retail pharmacies in opioid distribution and diversion (Janssen and Zhang, 2023; Gui et al., 2024), and studied the importance of physician characteristics, including training, practice style, beliefs about drug diversion, and market competition (Schnell and Currie, 2018; Staiger et al., 2022; Eichmeyer and Zhang, 2022; Currie et al., 2023; Schnell, 2025; Evans and Lieber, 2026). Place-based factors, such as the presence of “pill mill” pain clinics and regulatory policies, have also been shown to matter for risky opioid use (Finkelstein et al., 2025). On the demand side, individuals’ economic circumstances have been linked to risky opioid use (Carpenter et al., 2017; Hollingsworth et al., 2017; Case and Deaton, 2021). We contribute to this literature by connecting the supply and demand sides by highlighting the role of patient sorting in shaping opioid prescribing. We show that patient sorting

²Our counterfactual policy is in line with the DEA’s objective of reducing medically unjustified opioid prescriptions. To do so, the DEA aims to identify physicians who do not prescribe based on patients’ medical needs and revoke their license. However, it is rarely enforced in practice (Office of the Inspector General, 2019).

exacerbates observed variation in physician prescribing intensity and helps explain opioid misuse.

Next, this paper contributes to research on supply side policies that aim to limit access to opioids by studying the demand side response to such policies within the primary prescription market. A large body of research has studied the effects of various supply side policies such as Prescription Drug Monitoring Programs (Sacarny et al., 2016; Alpert et al., 2024; Sacarny et al., 2025). Some studies have shown that certain supply side restrictions invite demand side substitution to illicit drugs (Alpert et al., 2018; Evans et al., 2019; Soliman, 2025). We instead highlight a different demand-side response: patient resorting once physicians face restrictions, occurring within the prescription market. Resorting in the prescription market is important because exposure to prescription opioids facilitates transition into illicit opioid use (Eichmeyer and Zhang, 2022), with 80 percent of heroin users reporting prior prescription opioid misuse (Schnell, 2025). To study this mechanism, we develop an equilibrium model that jointly captures patient choice and physician prescribing. The model allows us to illustrate how patient sorting may affect the effectiveness of physician targeted policies and how to leverage sorting to improve such policies. Formally modeling sorting is novel; prior work has relied on settings with quasi-random assignment of patients to physicians, such as acute pain patients admitted through the Emergency Department or physician exit, to plausibly assume away sorting (Barnett et al., 2017; Eichmeyer and Zhang, 2022; Staiger et al., 2022).

Finally, we add to an emerging literature in health economics relying on two-way fixed effects estimators from the labor literature (for example, Abowd et al. (1999); Bonhomme et al. (2019)) to understand observed differences in care utilization or outcomes (Finkelstein et al., 2016; Molitor, 2018; Mourot, 2025). Finkelstein et al. (2016) use a “mover design” approach combined with a two-way fixed effects estimator to decompose the variance of health utilization into a patient component and a geographic area component. We contribute to the literature in two ways. First, we focus on more granular determinants of opioid prescribing behavior. In particular, we decompose the observed variance of physicians’ prescribing behavior into a patient component, a physician component, and patient sorting. Second, we show how patient sorting can exacerbate observed differences in physician prescription rates. Because we find evidence for strong positive sorting, implying that patients with stronger opioid preferences disproportionately seeing higher prescribing physicians, observed dispersion in prescription rates exceeds what differences in inherent physician prescribing intensity alone would produce.

2 Prescription Opioids for Chronic Pain

In this paper, we focus on the use of prescription opioids for the treatment of chronic pain in outpatient office visits. We first present the setting and afterwards discuss the data we use in our research.

2.1 Setting

We begin by discussing chronic pain in the United States. According to the Centers for Disease Control and Prevention (CDC), chronic pain is defined as an episode of pain that lasts for at least 3 months (CDC2016). Chronic pain affects a significant fraction of the general population. In 2023, around 24% of the United States’ adult population suffered from chronic pain (Lucas and Sohi, 2024). Due to its nature, chronic pain often leaves affected individuals incapacitated and unable to function properly without treatment. Treatment for chronic pain in the United States commonly involves the use of prescription opioids. Around 22% of US adults suffering from chronic pain rely on prescription opioids for pain management (Dahlhamer et al., 2021). Chronic pain is not a condition of old age and affects adults of all age groups. However, the use of prescription opioids for pain management is most common among working-age adults. According to the National Center for Health Statistics, adults aged 45-64 years exhibit the highest rates of prescription opioid use (Dahlhamer et al., 2021). Because most of the affected patients are of working age, almost 2 in 3 payments for opioid prescription collections are covered by private insurance (Schnell, 2025).

In the United States, patients must obtain prescription opioids by prescription from a licensed healthcare provider. About 4 in 5 prescription opioids in the United States are prescribed by physicians (Schnell, 2025). In outpatient settings, patients can generally choose the physician they want to see, with limitations on their choice set depending on their respective insurance network. Once a patient has chosen a physician, the physician examines the patient and decides whether to prescribe opioids or not. If prescribing opioids, the physician also has to decide on the number of days of supply. Contrary to acute pain, where initial supply in 33 states is limited to 5 to 7 days or less, no such regulation exists for opioids used for treating chronic pain (Department of Health and Human Services, 2021). After the days of supply are used, patients can get a refill if their prescription allows it, or they have to see a physician to obtain a new prescription.

Prescription opioids are a common part of pain management for both acute and chronic pain in the United States. The use of prescription opioids for *acute pain* for short periods of time is widely accepted as medically justified (CDC2016). However, the use of prescription opioids for *chronic pain* management is posing substantial health risks while lacking clear

evidence of any health benefits for patients (Volkow and McLellan, 2016). Metastudies have found an association of adverse health outcomes, such as increased risk of opioid abuse and dependence, all-cause mortality, and myocardial infarction, with long-term opioid treatment for chronic pain (Chou et al., 2020). There is also growing evidence that long-term prescription opioids may make it harder to effectively manage pain in chronic pain patients. Long-term opioid use may result in opioid induced hyperalgesia in patients, a condition that increases pain sensitivity that may extend beyond areas initially affected by chronic pain for which the opioids were prescribed (Lee et al., 2011). As a result, the recent CDC Clinical Practice Guideline for Prescribing Opioids for Pain only mentions four conditions that unequivocally warrant prescribing opioids for chronic pain: cancer-related pain treatment, sickle cell disease, palliative care, and end-of-life care (CDC2016).³ Outside of these conditions, prescription opioids should not be used as a first line of treatment, but the decision on whether to use prescription opioids for pain management is generally left to the physician and patient (CDC2016). This ambiguity leaves the eventual decision to use prescription opioids as a course of treatment up to the physicians’ discretion and patient preference.

2.2 MarketScan Data

As we discussed above, the majority of patients who receive prescription opioid treatment for chronic pain are privately insured working-age adults. For this reason, we need to assemble a data set that covers the relevant, privately insured working-age population comprehensively. Consequently, we cannot rely on the often-used data from the Centers for Medicare & Medicaid Services. Instead, we rely on data from the Merative MarketScan Research Databases. The MarketScan Commercial Database consists of individual-level, de-identified healthcare claims for millions of enrollees and their dependents in employer-sponsored healthcare plans in the United States. To construct our sample, we combine data from the Commercial Database and the Merative Micromedex RED BOOK for the years 2016 to 2022. We link the Outpatient Services Table, the Inpatient Services Table, and the Outpatient Pharmaceutical Claims Table from the Commercial Database with National Drug Code (NDC)-level information on opioids from the Merative Micromedex RED BOOK.

Using our data sources, we build two data sets for the New York City Metropolitan Statistical Area. We restrict our sample to patients who had an episode of chronic pain, received at least one opioid prescription between 2016 and 2022, and whose prescribing

³The 2016 CDC guideline marked a sharp pivot: in the pre-2016 era, following the medical profession’s recognition of pain as the “fifth vital sign” in the 1990s and the Joint Commission’s 2001 mandate to assess pain in all patients, high patient-reported pain scores were a strong trigger for opioid prescription (Mularski et al., 2006; Baker, 2017).

physician’s National Provider Identifier (NPI) is recorded in the data.⁴ The first data set consists of claims data information on all office visits of covered patients linked with data on their prescription collections. We refer to this data set as the *visit data set*. The second data set supplements these data with information on time periods in which a given patient chose not to see any physician. We refer to this data set as the *choice data set*. We use the former data set to establish patient sorting for our motivating evidence and in our structural estimation, and the second data set solely for our structural estimation.

Step I: Constructing the visit data set.

We construct our data sample in multiple steps. First, we develop a strategy to identify whether a particular office visit for a patient resulted in the prescription of a prescription opioid or not. Following the literature, we define an opioid prescription as any prescription of substances that fall within both, the opioid analgesics category under the American Hospital Formulary Service Classification Compilation Therapeutic Class scheme, and the Drug Enforcement Administration (DEA) drug schedule fall into schedule II or III, indicating high or moderate potential for abuse, respectively (Staiger et al., 2022; Alpert et al., 2024).⁵ As is common with commercial claims data, we lack NPI information for the prescribing physician for prescription opioid collections from the Outpatient Pharmaceutical Claims Table. Instead, we follow the literature to link prescription drug collections with prescribing providers. In particular, we follow Ding and Liu (2021) and assign a prescription to an observed office visit if the prescription opioid collection was within 14 days following the office visit, was the first collection within that time frame among all prescription opioid collections for that patient, and could not be associated with any prior office visit of that patient. Our final variable is a binary indicator for whether a prescription opioid was prescribed during a particular office visit. Note that we are focusing on the extensive rather than the intensive margin.

The MarketScan Commercial Database includes rich demographic and diagnosis information on possible patient risk factors. In particular, for each patient, we observe age, sex, and up to 4 primary diagnoses per outpatient visit. These diagnoses adopt the 10th revision of the International Classification of Diseases, ICD-10 codes. We leverage a patient’s medical history within 12 months preceding an office visit to create measures of a patient’s health status, as well as possible controls for opioid preference. The first health measure we construct to capture a wide range of comorbidities is the Charlson Comorbidity Index (Charlson et al.,

⁴We use the Chronic Conditions Warehouse (CCW) algorithm for “Chronic Pain”, revised February 2026.

⁵For the American Hospital Formulary Service Classification Compilation (AHFSCC) Therapeutic Class, the opioid analgesics category is “28:08.08”.

1987).⁶ Additionally, we follow the CDC Clinical Practice Guideline for Prescribing Opioids for Pain (CDC2016) to identify whether a patient has a medical condition unequivocally qualifying for prescription opioid treatment. These conditions are cancer-related pain, sickle cell disease, palliative care, and end-of-life care. Next, we construct possible controls for opioid preference by identifying diagnoses for which the CDC guidelines recommend against prescribing or urge caution when prescribing opioids. These conditions are sleep-disordered breathing, renal or hepatic insufficiency, mental health conditions (i.e., anxiety, depression, and post-traumatic stress disorder), substance use disorders, opioid abuse, and non-fatal overdose (CDC2016). We also construct counts of emergency room visits related to non-fatal overdoses, and count inpatient and outpatient visits during the 12 months preceding each office visit. We provide detailed descriptions, including the ICD-10 diagnosis codes used to construct our health status and opioid preference measures, in Appendices A and B.

Lastly, we construct measures of out-of-pocket costs for prescription opioids and total outpatient and inpatient spending during the 12 months preceding an office visit, using the financial variables from the MarketScan Commercial Database. Since we do not observe out-of-pocket costs when no opioid was prescribed, we construct out-of-pocket prescription opioid costs per outpatient visit as the mean out-of-pocket cost across insurance-type-month combinations. We provide more details on the construction of this variable in Appendix C. We summarize our first data set in table 1.

Step II: Constructing the choice data set.

For our choice data set, which we exclusively use to estimate our structural model, we supplement the above data with additional information on months when a patient could have seen a physician but chose not to see one.⁷ For each month, we define the population of active patients (the set of potential patients) as the set of patients who had at least one office visit in the corresponding quarter of that year. If we observe a patient visit a physician in a given month, we record their choice.⁸ If we do not observe a patient visiting a physician

⁶In particular, we use the 17 categories considered in the popular Charlson/Deyo variant (Deyo, 1992). We adopt the more recent weights suggested in Schneeweiss et al. (2003) and provide additional details in Appendix A.

⁷We use a month in accordance with our evidence on average days of supply.

⁸13% of patient-months involve more than one visit. For these cases, we collapse multiple visits into a single patient-month observation if they are all with the same physician (which occurs in about two-thirds of these cases), and we record an opioid prescription if any of the visits resulted in one. If a patient has multiple visits with different physicians, we retain the visit with the physician who has the highest number of prescriptions in that month across all patients. If there is a tie in the total number of prescriptions, we retain the visit with the physician who has the highest visit volume in that month across all patients.

Table 1: Summary Statistics of Chronic Pain Patients

Variable	Mean	SD	25 Perc	50 Perc	75 Perc
Panel A: Patient Severity and Constraints					
Age (years)	48.20	11.61	40.00	50.49	58.00
Sex (Male=1)	0.47	0.50	0.00	0.00	1.00
Charlson Comorbidity Index	1.23	1.59	0.00	1.00	2.00
Counts of Prior Inpatient Visits	0.13	0.40	0.00	0.00	0.00
Counts of Prior Outpatient Visits	65.26	70.60	23.50	45.50	83.40
Counts of Prior ED Visits	0.48	1.31	0.00	0.00	0.50
Prior OOP Spending	1696.09	2019.36	394.57	1065.30	2315.74
Out-of-pocket Cost	1.04	0.37	0.79	0.93	1.25
Panel B: Conditions Concordant with CDC Opioid Prescribing Guideline					
Sickle Cell Disease	0.00	0.07	0.00	0.00	0.00
Cancer-Related Pain	0.05	0.23	0.00	0.00	0.00
Panel C: Conditions Non-Concordant with CDC Opioid Prescribing Guideline					
Renal/Hepatic Deficiency	0.02	0.15	0.00	0.00	0.00
Mental Health Conditions	0.33	0.47	0.00	0.00	1.00
Opioid Misuse	0.04	0.21	0.00	0.00	0.00
Substance Use Disorder	0.12	0.33	0.00	0.00	0.00
Prior Nonfatal Overdose	0.03	0.18	0.00	0.00	0.00
Counts of ED Visits Related to Overdose	0.00	0.05	0.00	0.00	0.00
Work in Construction or Transportation	0.08	0.27	0.00	0.00	0.00
Observations					
Number of unique patients	11,909				

in a given month for which this patient is part of the active population, we record that the patient chose the outside option of not seeing a physician. If a patient chooses the outside option for a given month, we assume that their characteristics, which we computed for the first data set, are the same as the closest month in that quarter.

For our second data set, we also compute additional variables for physicians. Based on the choices of physicians or the outside option by patients, we compute the market shares for physicians for each month. The market share is defined as the number of patients we observe choosing a given physician in a given month, divided by the total number of patients active in that particular month. A common problem in defining market shares based on observable choices is that some options (physicians) may not be chosen in each month. If this happens for some of our physicians, we follow the standard approach in the discrete choice literature and assume that this physician was not active in a given month. We interpret this as the possibility that physicians might close their offices periodically for vacation. We summarize our data set for structural estimation in table 2.

Table 2: Physician Market Share Summary Statistics

Variable	Mean	SD	25 Perc	50 Perc	75 Perc
Physician Market Share (%)	0.09	0.09	0.04	0.06	0.09
Inside Share in a Market (%)	46.97	2.82	45.40	47.06	48.81
Observations					
Number of markets			84		
Number of physicians			8059		

3 Establishing Patient Sorting

In this section, we first document variation in opioid prescription rates for chronic pain patients across physicians. We then decompose the sources of this variation. A physician’s prescribing rate reflects not only their underlying prescribing propensity, but also differences in patient severity, patient preferences, and patient sorting into physicians based on both guideline-concordant and non-concordant preferences. We are particularly interested in assessing the relative importance of patient sorting versus physician prescribing propensity in explaining cross-physician differences in prescription rates. Finally, we extend the analysis beyond prescribing to opioid misuse, examining the extent to which patient sorting may contribute to risky prescription opioid use.

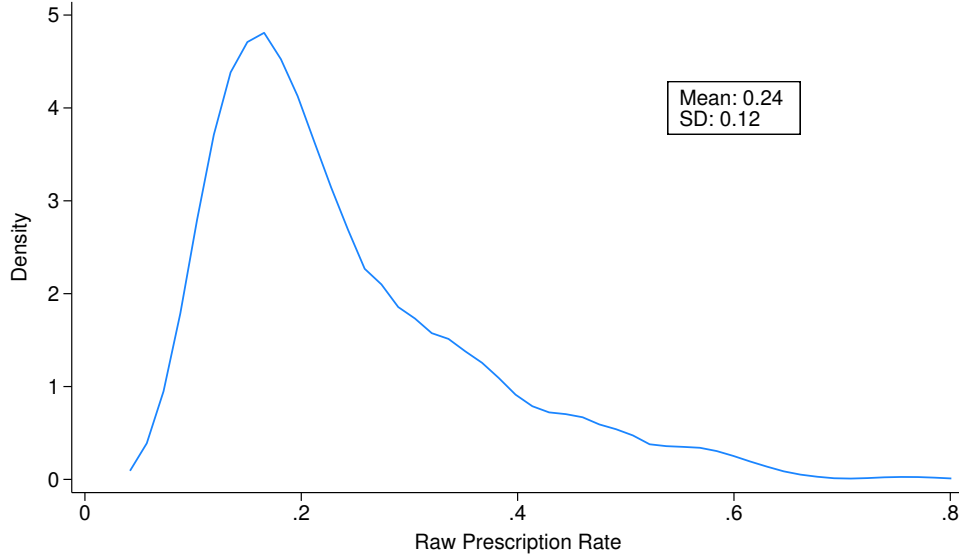
Cross-Physician Variation in Prescription Rates. Our primary outcome is an indicator equal to one if an office visit results in an opioid prescription. We estimate physician-level raw prescription rates by regressing this indicator on physician fixed effects:

$$\text{prescribe}_{it} = a_{j(i,t)} + \varepsilon_{it}, \quad (1)$$

where prescribe_{it} equals one if patient i receives an opioid prescription at visit t , $a_{j(i,t)}$ denotes the fixed effect for physician j , capturing physician j ’s average prescription rate in the data, and ε_{it} is the error term. Because $a_{j(i,t)}$ may be noisily estimated for physicians with relatively few patient visits, we apply an empirical Bayes shrinkage estimator that shrinks imprecisely estimated fixed effects towards the overall mean, following [Chandra et al. \(2016\)](#).

Figure 1 plots the density of the empirical Bayes-adjusted physician fixed effects, $\hat{a}_{j(i,t)}$. The distribution reveals substantial heterogeneity in raw prescription rates across physicians: the standard deviation is one-half of the mean prescription rate. This finding is consistent with prior evidence documenting large differences in drug prescribing intensity across providers ([Finkelstein et al., 2016](#); [Molitor, 2018](#); [Cutler et al., 2019](#); [Badinski et al., 2023](#); [Clemens et al., 2024](#)). The distribution also features a thick upper tail, indicating that a nontrivial share of physicians prescribe opioids at rates far above the average.

Figure 1: Raw Physician Prescription Rate



Notes: This figure plots the distribution of the empirical Bayes-adjusted physician fixed effects from equation (1). We restrict the sample to patients with at least 10 visits and physicians with at least 10 patient visits over the sample period.

Our results demonstrate that opioid prescription rates across physicians are highly dispersed. We next investigate the determinants of the observed dispersion in opioid prescription rates. A commonly believed source of opioid prescription rate dispersion may be differences in the patient severity composition across physicians. To assess whether opioid prescription rates mainly reflect differences in patient severity, we risk-adjust physicians' prescribing decisions by controlling for a rich set of patient observables. Our data features three categories of patient observables we leverage for risk-adjustment: (1) patient health status and patient-specific constraints, including patient age, sex, the Charlson comorbidity index, counts of historical inpatient, outpatient, and emergency-department visits, historical out-of-pocket spending, and insurance cost sharing; (2) patient medical conditions for which opioid use is clinically warranted under the CDC's opioid prescribing guidelines (CDC2016). These guideline-concordant conditions include sickle-cell disease and cancer-related pain; and (3) diagnoses and risk indicators for which the CDC recommends avoidance in the management of chronic pain. This group includes diagnoses of renal or hepatic insufficiency, mental health conditions, opioid misuse or substance-use disorder, a prior nonfatal overdose, counts of emergency-department visits related to overdoses, and an indicator for employment in construction or transportation industries.⁹

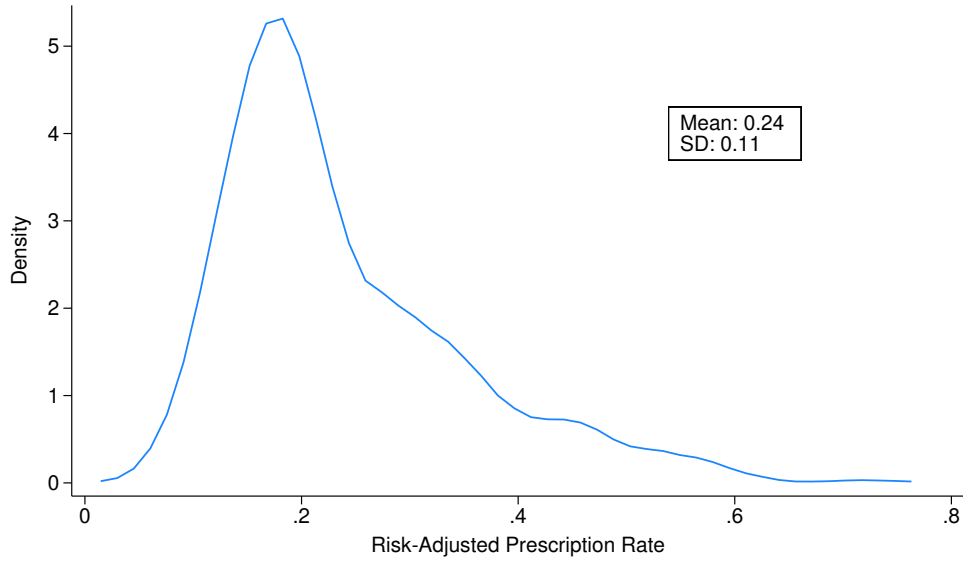
⁹Mental health conditions include anxiety, posttraumatic stress disorder, and depression (CDC2016). Occupations warranting caution include driving, using heavy equipment, climbing ladders, working at heights or around moving machinery, or working with high-voltage equipment (CDC2016).

We construct risk-adjusted prescription rates for each physician by regressing our prescription indicator on physician fixed effects, $a_{j(i,t)}$, the full set of patient observables described above, denoted by X_{it} , as well as year fixed effects, τ_t , to absorb any potential common time-varying factors such as policy changes and guideline updates using the following regression equation:

$$\text{prescribe}_{it} = a_{j(i,t)} + X_{it}b + \tau_t + \epsilon_{it}. \quad (2)$$

where $a_{j(i,t)}$ captures the physician-specific risk-adjusted mean prescription rate net of observable patient characteristics and common time effects. After applying the empirical Bayes shrinkage procedure to correct the estimated $\hat{a}_{j(i,t)}$ for potential measurement error, we plot the density for the empirical Bayes-adjusted $\hat{a}_{j(i,t)}$ in Figure 2.

Figure 2: Risk-adjusted Prescription Rate



Notes: This figure plots the distribution of the empirical Bayes-adjusted, risk-adjusted physician fixed effects from equation (2). We restrict the sample to patients with at least 10 visits and physicians with at least 10 patient visits over the sample period.

Figure 2 shows that relative to Figure 1, controlling for patient observables and year fixed effects leaves the dispersion in prescription rates largely intact. In other words, the prescription rate heterogeneity across physicians is not primarily driven by patient observables or common time trends. Specifically, the standard deviation of the distribution of risk-adjusted prescription rates is nearly identical to that of the raw prescription rates.

At face value, the patterns we documented appear to suggest a natural empirical rationale justifying policies that target high-prescribing physicians if all remaining variation in

risk-adjusted prescription rates reflects inherent differences in physicians’ prescribing propensities. However, such arguments and recommendations may be severely misguided. Observed prescription rates may reflect unobserved patient preferences and positive sorting of patients with a strong preference for prescription opioids into high-prescribing physicians, even after risk-adjustment for patient severity. In fact, positive sorting may amplify even modest differences in physicians’ underlying prescribing propensities and lead to vast dispersion in observed prescribing rates. Conversely, if patient sorting were negative, the true dispersion in physician prescribing propensities may be larger than the one implied by observed risk-adjusted prescription rates.

Patient Sorting. To understand the importance of patient sorting versus physicians’ prescribing propensities for the observed dispersion in prescription rates, we exploit the panel structure of our data. In particular, we decompose the variation in raw prescription decisions into components attributable to patient type, physician type, and sorting of patients into physicians whose prescribing styles they prefer. We further distinguish sorting based on guideline-concordant versus non-concordant preferences.

Our variance decomposition approach mirrors the two-way fixed effects estimator of [Abowd et al. \(1999\)](#) from the labor literature, but applied to prescription decisions. Following their method, we specify a two-way fixed effects regression in which we regress physicians’ prescription decisions on patient fixed effects, physician fixed effects, patient observables, and year dummies:

$$\text{prescribe}_{it} = \eta_{j(i,t)} + \theta_i + X_i\beta + \tau_t + \varepsilon_{it}, \quad (3)$$

where $\eta_{j(i,t)}$ represents the physician prescribing propensity of the chosen physician, θ_i represents patient unobserved type, and X_i is a vector of patient observables defined as above, τ_t are year dummies, and ε_{it} is an idiosyncratic error term representing any unobserved time-varying patient characteristics.¹⁰

Note that in equation (3), β and θ_i are not separately identified because θ_i and X_i are both patient-specific. Therefore, we define $\Theta_i \equiv \theta_i + \beta X_i$ and treat Θ_i as patient fixed effect for estimation in the spirit of [Abowd et al. \(1999\)](#). We therefore rewrite equation (3) as

¹⁰Because most of patient observables exhibit little within-patient variation over time, we use patient-level averages and treat them as time-invariant characteristics. Including time-varying versions of these variables when available yields highly similar results in terms of the relative importance of each component in explaining variation in prescription rates.

$$\begin{aligned}
\text{prescribe}_{it} &= \eta_{j(i,t)} + \underbrace{\theta_i + X_i\beta}_{\equiv \Theta_i} + \tau_t + \varepsilon_{it} \\
&= \eta_{j(i,t)} + \Theta_i + \tau_t + \varepsilon_{it}.
\end{aligned} \tag{4}$$

Obtaining unbiased fixed effects in an [Abowd et al. \(1999\)](#) style model is challenging because physician fixed effects are identified only from patients who switch physicians. When switching is limited, fixed effects estimates can be severely biased, a problem known as limited mobility bias ([Andrews et al., 2012](#); [Bonhomme et al., 2023](#)). Recent advances in the labor literature show that this bias can be mitigated using a two-step estimator that first groups firms and then estimates group-level fixed effects ([Bonhomme et al., 2019](#)). We adopt the approach of [Bonhomme et al. \(2019\)](#) and follow [Mourot \(2025\)](#) to adapt this approach to our setting of opioid prescribing. Following [Mourot \(2025\)](#), in the first stage, we group physicians and patients using k-means clustering based on physician-level and patient-level risk-adjusted prescription rates, respectively.¹¹ Appendix [D](#) provides more details on the grouping algorithm. Following standard practice in the labor literature ([Kline, 2024](#)), we use ten groups for physicians and ten groups for patients. We assess robustness by varying the number of groups. In the second stage, we estimate equation (4) using group-level physician and patient fixed effects.

After obtaining estimates for the group-level fixed effects, $\hat{\eta}_{j(i,t)}$ and $\hat{\Theta}_i$, we further decompose $\hat{\Theta}_i$ into guideline-concordant and non-concordant preference components. To do so, we regress patient fixed effects estimates, $\hat{\Theta}_i$, on our full vector of non-time-varying patient observables X_i :

$$\hat{\Theta}_i = \underbrace{X_{1i}\beta_1}_{\text{guideline-concordant}} + \underbrace{X_{2i}\beta_2 + \theta_i}_{\text{guideline-non-concordant}}. \tag{5}$$

We partition X_i into two sets of time-invariant patient characteristics: guideline-concordant conditions, X_{1i} , and guideline nonconcordant preferences, X_{2i} . X_{1i} includes diagnoses that warrant opioid use, including sickle-cell disease and cancer-related pain. We collect the remaining guideline non-concordant factors in X_{2i} and θ_i . X_{2i} includes health severity and patient-specific constraints, as defined before, and characteristics for which the CDC explicitly recommends avoidance of opioid use for managing chronic pain. θ_i is the residualized patient fixed effects obtained after partialing out X_i from $\hat{\Theta}_i$.

¹¹[Mourot \(2025\)](#) groups surgeons and hospitals, whereas we group patients and physicians.

We now decompose the variance of the prescribing decision. Following the recommendation of Kline (2024), we scale each component of the decomposition by $var(\text{prescribe}_{it}^E)$, the variation in prescribing that can be explained by either the patient or physician side factors. According to Kline (2024), this normalization ensures the comparability of our results with any future related studies. In particular, we have

$$\begin{aligned} var(\text{prescribe}_{it}^E) &\equiv var(\eta_{j(i,t)} + X_{1i}\beta_1 + X_{2i}\beta_2 + \theta_i) \\ &= var(\eta_{j(i,t)}) + var(X_{1i}\beta_1) + var(X_{2i}\beta_2 + \theta_i) \\ &\quad + 2cov(X_{1i}\beta_1, \eta_{j(i,t)}) + 2cov(X_{2i}\beta_2 + \theta_i, \eta_{j(i,t)}) + 2cov(X_{1i}\beta_1, X_{2i}\beta_2 + \theta_i) \end{aligned} \quad (6)$$

where $var(X_{1i}\beta_1)$ captures variation in prescribing explained by guideline-concordant patient preferences, while $var(X_{2i}\beta_2 + \theta_i)$ reflects variation associated with guideline-non-concordant patient preferences. The term $var(\eta_{j(i,t)})$ represents variation attributable to physician prescribing propensity. The covariance term $cov(X_{1i}\beta_1, \eta_{j(i,t)})$ captures patient sorting based on guideline-concordant preferences, whereas $cov(X_{2i}\beta_2 + \theta_i, \eta_{j(i,t)})$ captures sorting based on guideline-non-concordant preferences. Finally, $cov(X_{1i}\beta_1, X_{2i}\beta_2 + \theta_i)$ reflects the covariance between guideline-concordant and non-concordant preferences. Therefore, $var(X_{1i}\beta_1) + var(X_{2i}\beta_2 + \theta_i) + 2cov(X_{1i}\beta_1, X_{2i}\beta_2 + \theta_i)$ represents the portion of variation captured by the patient type Θ_i . We report the results of our variance decomposition in Table 3.

For our main results in Table 3, the baseline specification, we use 10 groups for both patients and physicians, and restrict the sample to patients with at least 2 visits and physicians with at least 2 patient visits over the sample period. We report results from alternative sample restrictions and alternative numbers of groups to assess the robustness of our results. Additionally, we report the results of employing an empirical Bayes-adjusted estimator for the patient and physician fixed effects.

Our results in Table 3 demonstrate that across specifications, and irrespective of whether we apply Bayesian shrinkage, there exists strong positive assortative matching between patients and physicians.¹² We find that assortative matching is overwhelmingly due to guideline-non-concordant preferences for opioid prescriptions. Patient sorting based on guideline-concordant preferences explains only about 2% of the variation in prescription rates, whereas sorting based on non-concordant preferences accounts for nearly 30%. Perhaps surprisingly, physicians' intrinsic prescribing propensity explains only around 11% of

¹²Due to space constraints, we do not report the term $2cov(X_{1i}\beta_1, X_{2i}\beta_2 + \theta_i)$, which is close to zero. As a result, summing the first five rows for a given column yields nearly 100%.

the variation in prescribing decisions. Our results imply that patient sorting accounts for roughly three times as much of the observed heterogeneity in prescribing rates as providers' own prescribing propensity. Consistent with this finding, the correlation coefficient between patients' non-concordant preferences and their chosen physicians' prescribing propensity is as high as 0.55.¹³

¹³In practice, some medical consensus exists on managing chronic pain beyond the two conditions concordant with the CDC guidelines. That is, physicians may also consider observable patient characteristics related to health status, including age, sex, the Charlson comorbidity index, counts of historical inpatient, outpatient, and emergency-department visits, and historical out-of-pocket spending, as well as unobserved factors such as pain severity. In our analysis, the observable characteristics reflecting health status explain only 1% of the patient fixed effects, and patient sorting is not driven by these factors. The majority of sorting therefore remains based on factors concordant with neither the guidelines nor broader medical consensus. To the extent that some medical consensus reflects compassionate prescribing and may be viewed differently from preferences such as opioid addiction, our counterfactual policies deliver a range of estimates depending on whether physicians can prescribe based on part or all of the medical consensus.

Table 3: Percentage of Variance Explained in Raw Prescription Rates (%)

	Baseline	Alternative Number of Groups				Alternative Sample Restrictions, $K = 10$			Bayesian Shrinkage
	$K = 10$ ≥ 2 visits/patient ≥ 2 visits/provider	$K = 5$	$K = 15$	$K = 20$	$K = 50$	≥ 1 visits/patient ≥ 1 visits/provider	≥ 5 visits/patient ≥ 5 visits/provider	≥ 10 visits/patient ≥ 10 visits/provider	$K = 10$ ≥ 10 visits/patient ≥ 10 visits/provider
Patient									
$\frac{\text{Var}(X_{2i}\beta_2 + \theta_i)}{\text{Var}(\text{prescribe}^E)}$	58.36	58.69	58.49	58.61	59.12	56.41	59.61	60.14	63.46
$\frac{\text{Var}(X_{1i}\beta_1)}{\text{Var}(\text{prescribe}^E)}$	1.88	1.81	1.87	1.80	1.80	1.43	3.38	3.49	3.66
Provider									
$\frac{\text{Var}(\eta_{j(i,t)})}{\text{Var}(\text{prescribe}^E)}$	10.99	10.82	11.00	10.94	10.95	12.87	8.45	7.42	5.96
Sorting									
$2 \times \frac{\text{Cov}(X_{2i}\beta_2 + \theta_i, \eta_{j(i,t)})}{\text{Var}(\text{prescribe}^E)}$	28.03	27.77	27.91	27.90	27.41	28.67	27.26	27.30	25.46
$2 \times \frac{\text{Cov}(X_{1i}\beta_1, \eta_{j(i,t)})}{\text{Var}(\text{prescribe}^E)}$	0.72	0.89	0.72	0.72	0.70	0.58	1.27	1.53	1.41
Correlation									
$\text{corr}(X_{2i}\beta_2 + \theta_i, \eta_{j(i,t)})$	0.55	0.55	0.55	0.55	0.54	0.53	0.61	0.65	0.65
$\text{corr}(X_{1i}\beta_1, \eta_{j(i,t)})$	0.08	0.10	0.08	0.08	0.08	0.07	0.12	0.15	0.15

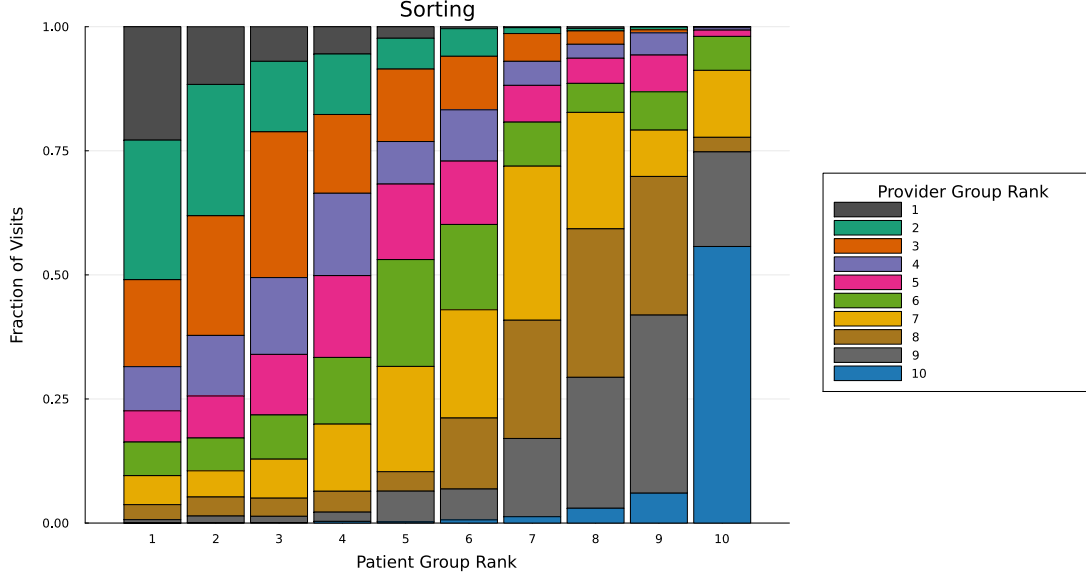


Figure 3: Patient Sorting

We further illustrate strong positive assortative matching in Figure 3. Figure 3 shows that patients with low opioid preference (patient group 1) make more than half of their visits to providers who prescribe the least opioids (provider groups 1 and 2). Conversely, patients with high opioid preference (patient group 10) disproportionately choose providers with the highest prescription rates (provider group 10). Figure 14 in Appendix E shows the same patterns when patients and providers are divided into 5 groups instead of 10.

Positive patient sorting implies that the observed differences in prescription rates across physicians are larger than the underlying differences in physicians’ true prescribing propensities. In other words, positive sorting amplifies variation in prescribing behavior, leading to a steeper gradient in the raw prescription rate. Figure 4 illustrates this mechanism: the horizontal axis represents ten physician groups ranked by their raw prescription rates. Each blue triangle shows the raw prescription rate for a physician group, while each red circle indicates the estimated physician fixed effect, i.e., the group’s prescribing propensity, scaled by a base prescription rate. The fitted line for the raw prescription rate is visibly steeper, reflecting positive patient sorting. At the upper end of the distribution, physicians appear to prescribe much more liberally than they truly are: raw prescription rates overstate their true prescribing propensities. Similarly, physicians in the lower percentiles appear to prescribe far more conservatively, even though their true propensities are not that different from those of higher prescribers.

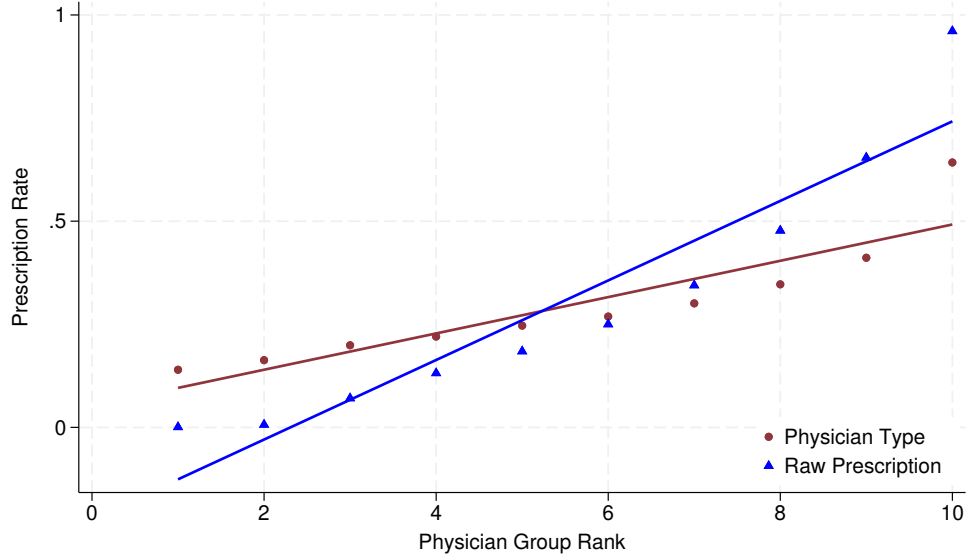


Figure 4: Patient Sorting Versus Physician Propensity Type

Patient Sorting Contributes to Risky Opioid Use. We have established that variation in observed cross-physician prescription rates is driven less by differences in physicians’ prescribing propensities and more by patients sorting into physicians according to those propensities. We now examine whether positive patient sorting also plays an important role in explaining adverse health outcomes. To do so, we revisit the decomposition exercise in equation (4), replacing the prescription outcome variable with an indicator for opioid misuse at that visit. Literature has documented a correlation between high prescription levels, defined as the patient’s daily MED being 120 mg or greater, and opioid abuse (Meara et al., 2016; Staiger et al., 2022; Finkelstein et al., 2025). Following this literature, we classify a visit as involving opioid misuse if the patient’s daily MED is 120 mg or greater. We provide details on the construction of our measure of daily MED in Appendix F. We focus on this health outcome because it is well defined at the visit level. Additionally, it provides a more objective measure of opioid misuse than opioid-abuse-related diagnoses coded by the physician, which depend on both the physician’s or patient’s willingness to test and assessment. Moreover, opioid misuse is both a strong risk factor for, and a precursor to, opioid addiction. Based on this definition, 4% of visits in our sample involve risky opioid use.¹⁴

In Table 4, we decompose the explained variation in opioid misuse into components attributable to guideline-concordant patient preferences ($X_{1i}\beta_1$), guideline-non-concordant

¹⁴Finkelstein et al. (2025) used data from the Social Security Disability Insurance (SSDI) in a 20% random sample of Medicare beneficiaries from 2006 through 2019 and identified 2.98% of visits involve risky opioid use.

patient preferences ($X_{2i}\beta_2 + \theta_i$), physician prescribing propensity ($\eta_{j(i,t)}$), and patient sorting on both guideline-concordant and non-concordant preferences ($Cov(X_{1i}\beta_1, \eta_{j(i,t)})$ and $Cov(X_{2i}\beta_2 + \theta_i, \eta_{j(i,t)})$). All specifications yield qualitatively similar results. Moreover, the patterns closely mirror those from the decomposition of raw opioid prescription rates. Patient sorting explains nearly one-third of the variation in opioid misuse, far outweighing the contribution of physician prescribing propensity, which accounts for less than 10% in the baseline specification. The correlation coefficient between patient non-concordant preference type and physician propensity is as high as 0.54. Furthermore, the contribution of guideline-concordant preference is much smaller in this context, suggesting that non-concordant preferences play a central role in driving opioid misuse. These results indicate that patient sorting not only drives high opioid prescription rates but is also associated with worse health outcomes, thereby amplifying the opioid epidemic.

Table 4: Percentage of Variance Explained in Opioid Misuse (%)

	Baseline	Alternative Number of Groups				Alternative Sample Restrictions, $K = 10$			Bayesian Shrinkage
	$K = 10$ ≥ 2 visits/patient ≥ 2 visits/provider	$K = 5$	$K = 15$	$K = 20$	$K = 50$	≥ 1 visits/patient ≥ 1 visits/provider	≥ 5 visits/patient ≥ 5 visits/provider	≥ 10 visits/patient ≥ 10 visits/provider	$K = 10$ Bayes shrinkage
Patient									
$\frac{\text{Var}(X_{2i}\beta_2 + \theta_i)}{\text{Var}(\text{opioid misuse}^E)}$	60.93	57.87	64.05	64.10	64.26	57.83	62.92	57.52	66.23
$\frac{\text{Var}(X_{1i}\beta_1)}{\text{Var}(\text{opioid misuse}^E)}$	0.98	1.20	0.93	0.93	0.91	0.77	1.40	1.49	1.64
Provider									
$\frac{\text{Var}(\eta_{j(i,t)})}{\text{Var}(\text{opioid misuse}^E)}$	10.42	12.85	8.49	8.52	8.46	12.41	8.25	10.28	6.15
Sorting									
$2 \times \frac{\text{Cov}(X_{2i}\beta_2 + \theta_i, \eta_{j(i,t)})}{\text{Var}(\text{opioid misuse}^E)}$	27.37	27.71	26.25	26.15	26.07	28.74	26.96	30.12	25.40
$2 \times \frac{\text{Cov}(X_{1i}\beta_1, \eta_{j(i,t)})}{\text{Var}(\text{opioid misuse}^E)}$	0.34	0.43	0.35	0.35	0.35	0.30	0.50	0.56	0.58
Correlation									
$\text{corr}(X_{2i}\beta_2 + \theta_i, \eta_{j(i,t)})$	0.54	0.51	0.56	0.56	0.56	0.54	0.59	0.62	0.63
$\text{corr}(X_{1i}\beta_1, \eta_{j(i,t)})$	0.05	0.05	0.06	0.06	0.06	0.05	0.07	0.07	0.09

Threats to Identification. Including patient fixed effects, which absorb all time-invariant unobserved patient characteristics, addresses many sources of possible confounders. However, fixed effects estimates may still be biased if patients’ switching decisions are correlated with unobserved time-varying characteristics ε_{it} . Following the labor literature, we provide evidence that switching is not driven by omitted time-varying unobservables in our setting. In particular, [Card et al. \(2018\)](#) show that if worker switching is exogenous, workers exhibit symmetric wage responses with opposite signs when switching to higher- versus lower- type firms. Analogously, if patient switching is exogenous in our context, patients who switch to more aggressive prescribers should experience prescription probability changes that are symmetric, with opposite signs, to those who switch to less aggressive prescribers.

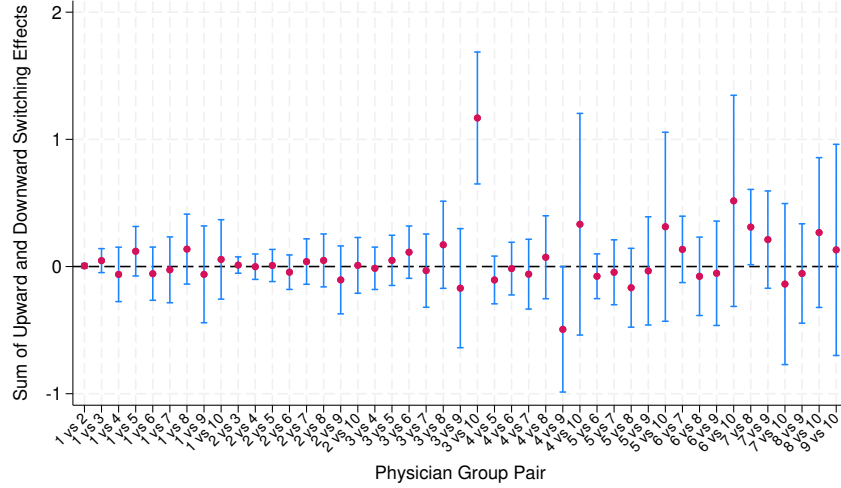
We test this implication in Figure 5. Figure 5 presents pairwise analyses of movers between physician groups. For each pair of physician groups, we estimate the effect of switching across groups relative to remaining within the same group. For example, for the physician group pair 1 and 2, some patients move from physician group 1 to group 2 (“move up”), others move from group 2 to group 1 (“move down”), while patients who switch physicians but remain within the same group serve as a control group. We implement a difference-in-differences-style design to estimate the effect of crossing groups relative to staying within a group.

Across all 45 group pairs, with one exception, we find that the change in prescription probability has opposite signs for move-up and move-down patients, and that the magnitudes are highly similar. As shown in Figure 5, each dot reports the sum of the estimated move-up and move-down effects, which is generally close to zero and statistically insignificant. This symmetric response pattern holds across almost all pairs of physician groups.

We further conduct an event-study analysis, pooling all movers into move-up and move-down categories, to examine whether switching is preceded by differential pre-trends. As shown in Figure 6, we find no evidence of pre-trends in prescribing behavior prior to the switch. This absence of pre-trends further suggests that switching is not driven by patients’ gradual learning or other time-varying unobserved factors.

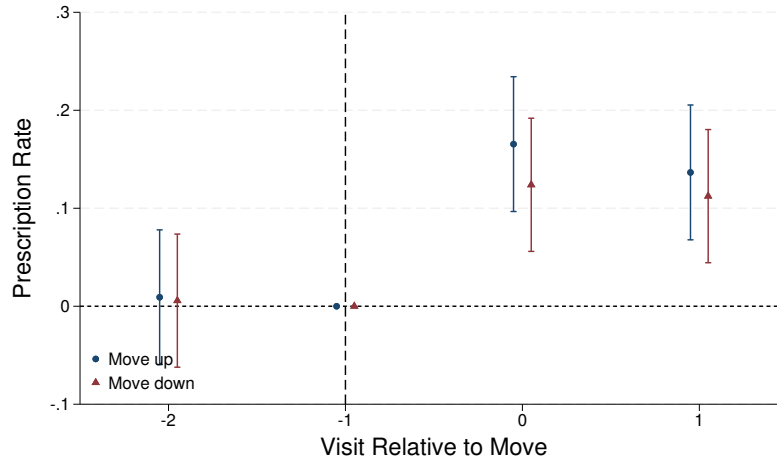
Discussion. Our finding that patient sorting dominates providers’ prescribing propensity in explaining variation in both prescription rates and opioid misuse has important policy implications. First, policies that target only high prescribers are unlikely to be effective in curbing prescription rates or opioid misuse, as patients who initially sought care from these top prescribers are likely to resort to other providers whose prescribing propensity is just below that of the targeted prescribers. Second, exploiting patient sorting in physician-targeted policies might be a promising avenue to address opioid over-prescription, especially

Figure 5: Net Prescription Rate Response to Upward and Downward Patient Switching



Notes: This figure plots the sum of the changes in prescription rates for patients who switch upward and downward within each pair of physician groups. Physician groups are defined based on physician-level risk-adjusted prescription rates and are ordered from 1 (lowest-prescribing group) to 10 (highest-prescribing group).

Figure 6: Magnitude of Prescription Rate Responses to Patient Switching



Notes: This figure plots the magnitude of changes in prescription rates for patients who switch to higher-prescribing physicians (“move up”) and to lower-prescribing physicians (“move down”). Time 0 corresponds to the visit at which the physician switch occurs.

if policy can discourage patient sorting based on guideline-non-concordant preferences while preserving patient sorting based on guideline-concordant conditions.

To examine how patient sorting interacts with top-prescriber-targeted policies and to evaluate alternative policies, we propose and estimate a structural model that explicitly accounts for patient sorting in the next section.

4 Model

Our results in section 3 suggest that patients choose physicians according to physicians' prescribing propensities. In this section, we develop a static two-stage model of physician-patient interaction that is able to capture the uncovered patient sorting. In the first stage, the *choice stage*, in each month t , patients choose either to see a physician $j \in \mathcal{J}_t$ in their market (month) or to select the outside option of not seeking care. Patients choice is based on their beliefs about each physician's prescribing behavior. In the second stage, the *prescription stage*, the chosen physician decides whether to prescribe or not. We illustrate our two-stage model for a single month t and a single patient i in figure 7.

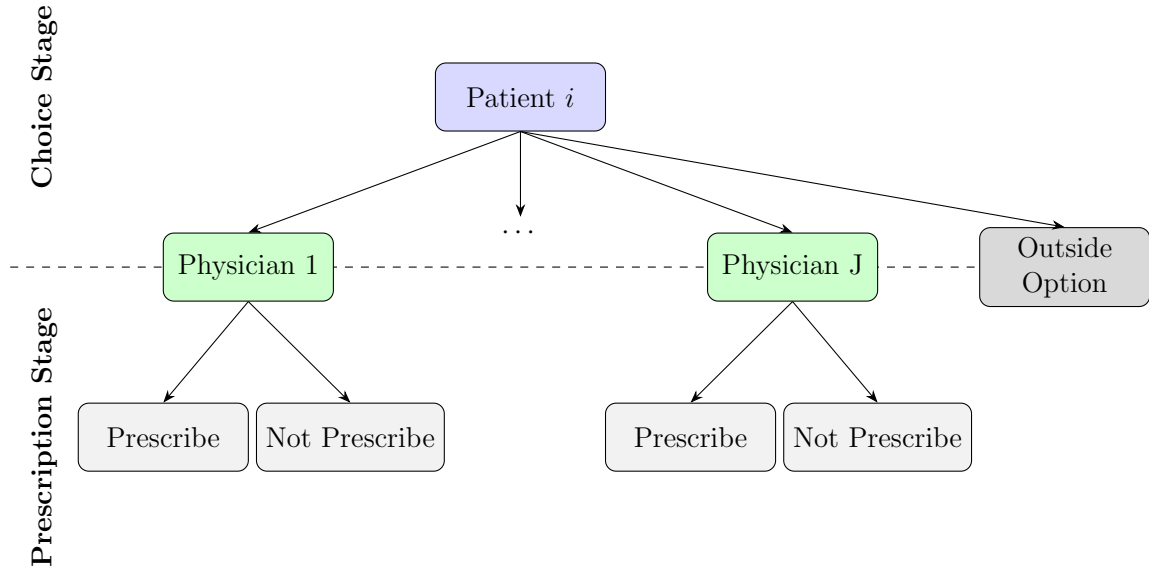


Figure 7: Illustration of two-stage model for each month t .

4.1 Indirect Utility and Information Environment

Physician's Indirect Utility Function. In each month t , conditional on being chosen by patient i , physician j makes a binary treatment decision: prescribe an opioid or not.

Physician j 's indirect utility from treating patient i is given by

$$V_{ijt} = -\gamma + \pi_{jt} + \eta_j a_{ijt} + U_{ijt}(a_{ijt}) + \varepsilon_{ijt}(a_{ijt}). \quad (7)$$

Here, π_{jt} represents the baseline profit from seeing a patient, measured in utils and independent of treatment choices, such as the office visit fee. η_j is a physician-specific fixed effect capturing physician j 's inherent propensity to prescribe opioids, reflecting factors such as treatment philosophy, graduate medical education, risk tolerance, and regulatory scrutiny. a_{ijt} equals 1 if physician j prescribes an opioid to patient i in month t , and 0 otherwise. $U_{ijt}(a_{ijt})$ represents patient's utility. γ represents Euler's constant, and ε is an i.i.d. alternative-specific Type 1 extreme value shock, independent of all other components. In this specification, everything else equal, physicians with higher η_j are more likely to prescribe, whereas those with lower η_j prescribe less often, *independent* of the patient's utility from receiving a prescription. For estimation, we impose that one of the η_j parameters of our model is equal to 0.

Physicians' Information. Physicians make prescription decisions by combining patient preferences that they can learn with their own treatment preferences. While physicians are medical experts, they may not fully observe a patient's preferences. We assume that the components of patient preference that are observable and unobservable to the physician are additively separable, such that

$$U_{ijt}(a_{ijt}) = u_{ijt}(a_{ijt}) + \nu_{ijt}, \quad (8)$$

where $u_{ijt}(a_{ijt})$ represents the part of patient utility observable to the physician, and ν_{ijt} the unobservable part. We assume that while physicians do not observe the realization of ν_{ijt} , they know its distribution. In addition, we assume that this informational structure is common knowledge to both patients and physicians.

Patient's Indirect Utility Function. We start by defining patient types in our model. We assume that patient i 's type is a tuple (Θ_i, ζ_i) , consisting of two time-invariant components. The first component, Θ_i , captures both guideline-concordant and non-concordant preferences for opioids. The second component, ζ_i , measures how informed patient i is about physicians' prescription types. We assume that Θ_i enters patients' utility, whereas the information type ζ_i affects patient choice only through patient beliefs and does not directly enter utility. We refer to Θ_i as the patient's *opioid preference type* and to ζ_i as the patient's *information type*.

Patient i 's indirect utility from choosing physician j at time t is

$$U_{ijt}(a_{ijt}) = \underbrace{\delta_{jt} + a_{ijt}(\Theta_i + \tau_t)}_{\equiv u_{ijt}(a_{ijt})} + Z_{it}\phi + \nu_{ijt}, \quad (9)$$

where δ_{jt} denotes the mean utility associated with physician j in month t . The mean utility varies across physicians because they differ in amenities or are subject to physician-specific demand shocks. For example, a physician may have more friendly front desk staff, offer appointment reminders, or be perceived as having greater expertise in treating chronic pain, independent of their prescribing behavior.

Θ_i measures patient i 's inherent preference for opioid prescriptions. As discussed in section 3, it includes both patient guideline-concordant and non-concordant preferences, which may arise from prior opioid misuse or addiction. As in equation (5), we assume that patients' preference types are a linearly separable function of patients' guideline-concordant needs ($X_{1i}\beta_1$) and non-concordant preference ($X_{2i}\beta_2 + \theta_i$).

τ_t denotes a yearly fixed effect for patient taste for prescription opioids. Z_{it} represents time-varying patient characteristics that affect the utility from an office visit at time t , and ν_{ijt} is an i.i.d. Type 1 extreme value shock. We further assume that the distribution of patients' opioid preference type Θ_i has mean zero, $E[\Theta_i] = 0$, and normalize the value of the outside option (not seeing any physician) to $U_{i0t} = \nu_{i0t}$.

Patients' Information. We assume that patients do not have perfect information about physicians' preferences. First, patients cannot observe the alternative-specific preference shock $\varepsilon_{ijt}(a_{ijt})$, because this shock is realized only after a physician is chosen. We assume, however, that patients know the distribution of this shock. Second, patients do not perfectly observe physicians' inherent prescription propensity η_j when choosing a physician. Instead, we assume that patient i 's perceived prescription type of physician j , denoted by ω_{ij} , equals the physician's true type plus noise:

$$\omega_{ij} = \eta_j + \zeta_{ij}. \quad (10)$$

A patient's information type ζ_{ij} captures their uncertainty about a physician's prescription type η_j . We allow patients to differ in their level of informedness for each physician. In practice, patients may differ in their ability to search for and interpret informal information sources, such as online forums (e.g., Bluelight, Reddit), word-of-mouth, or prior experiences, which can systematically shape how well informed they are about physicians' willingness to prescribe opioids.

We further assume that $\zeta \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_\zeta^2)$. Under this information structure, a patient may overestimate one physician's prescription propensities and at the same time underestimate another physician's. We interpret this assumption as reflecting idiosyncratic misperceptions about individual physicians, depending on the amount and source of information a patient has for different physicians. Our assumptions imply that these misperceptions share a common variance σ_ζ^2 . If all patients have an accurate perception of each physician's prescribing type, the variance of ζ_{ij} will be zero. Finally, we assume that the information structure of the game is common knowledge to both physicians and patients.

Given the two sources of uncertainty regarding a physician's preference, patients cannot perfectly predict a physician's treatment decision at the time of choosing a physician. Instead, they form expectations about a physician's probability of prescribing based on their perceived prescription type.

4.2 Equilibrium

This two-stage model is solved by backward induction. We start by describing a physician's optimal prescribing behavior, conditional on being chosen by the patient. We then derive patients' optimal physician choice in the first stage.

4.2.1 Optimal Prescription Choice

Because the physician can observe the patient's utility up to the realization of the shock ν_{ijt} , we can rewrite the physician's indirect utility V_{ijt} as an explicit function of the prescription decision a_{ijt} :

$$V_{ijt}(a_{ijt} = 1) = \pi_{jt} + \eta_j + \delta_{jt} + \Theta_i + \tau_t + Z_{it}\phi + \varepsilon_{ijt}(1), \quad \text{and} \quad (11)$$

$$V_{ijt}(a_{ijt} = 0) = \pi_{jt} + \delta_{jt} + Z_{it}\phi + \varepsilon_{ijt}(0). \quad (12)$$

Physician j will choose to prescribe if her indirect utility from prescribing an opioid weakly exceeds the indirect utility from not prescribing i:

$$V_{ijt}(a_{ijt} = 1) - V_{ijt}(a_{ijt} = 0) = \eta_j + \Theta_i + \tau_t + \varepsilon_{ijt}(1) - \varepsilon_{ijt}(0) \geq 0. \quad (13)$$

Note that this difference and, therefore, the physician's prescription decision, is *independent* of the demand side parameters δ_{jt} , the patient's information type ζ_i , and the base profit from treating a patient π_{jt} . Therefore, in deciding whether to prescribe, the physician does

not consider her relative attractiveness to the patient or the information a patient has about her prescription type in the first stage. Instead, the prescription decision depends solely on the patient's preference from receiving the prescription and the physician's own preference for prescribing opioids.

Since we assume that the alternative-specific error terms ε follow type 1 extreme value distribution, the probability that physician j prescribes an opioid to patient i at time t can be expressed as:

$$\rho_{ijt} = \frac{\exp(\eta_j + \Theta_i + \tau_t)}{1 + \exp(\eta_j + \Theta_i + \tau_t)}. \quad (14)$$

4.2.2 Optimal Physician Choice

Patients face a discrete choice among J_t physicians in a market, given their information about each physician. As described in section 4.1, patients do not observe the realization of the type 1 extreme value shocks ε and instead hold potentially inaccurate perception ω_{ij} about each physician's prescription propensity η_j . Based on this information, patients form beliefs over the probability of receiving a prescription from physician j :

$$\tilde{\rho}_{ijt} = \frac{\exp(\omega_{ij} + \Theta_i + \tau_t)}{1 + \exp(\omega_{ij} + \Theta_i + \tau_t)}. \quad (15)$$

Given the patient's beliefs about physician j 's prescribing probability, $\tilde{\rho}_{ijt}$, conditional on being chosen, and the assumption that ν_{ijt} follows a Type 1 extreme value distribution, the probability that patient i chooses physician j at time t can be expressed as:

$$s_{ijt} = \frac{\exp(\delta_{jt} + \tilde{\rho}_{ijt} \times (\Theta_i + \tau_t) + Z_{it}\phi)}{1 + \sum_k \exp(\delta_{kt} + \tilde{\rho}_{ikt} \times (\Theta_i + \tau_t) + Z_{it}\phi)}. \quad (16)$$

Lastly, we obtain aggregate choice probabilities s_{jt} by integrating over individual choice probabilities s_{ijt}

$$s_{jt} = \int s_{ijt} di. \quad (17)$$

5 Identification

The parameter vector of the model is $\{\boldsymbol{\eta}, \boldsymbol{\Theta}, \boldsymbol{\tau}, \boldsymbol{\delta}, \sigma_\zeta, \boldsymbol{\phi}, \boldsymbol{\pi}\}$. We start by noting that physicians' base profit from treating patients, π_{jt} , is neither identified nor relevant for physicians'

prescribing decisions, because only differences in utilities matter, as is standard in discrete choice models. Therefore, without loss of generality, we normalize π_{jt} to zero.

For the remaining parameters, our identification strategy exploits the fact that the parameter vector can be partitioned into two sets. The first set consists of parameters that govern both the choice stage and the prescription stage of the model, $\Xi_1 = \{\boldsymbol{\eta}, \boldsymbol{\Theta}, \boldsymbol{\tau}\}$. The second set consists of parameters that exclusively govern the choice stage, $\Xi_2 = \{\boldsymbol{\delta}, \sigma_\zeta, \boldsymbol{\phi}\}$. This partition allows us to separately discuss the sources of variation identifying Ξ_1 and Ξ_2 .

We start with the first set of parameters, $\Xi_1 = \{\boldsymbol{\eta}, \boldsymbol{\Theta}, \boldsymbol{\tau}\}$, which govern both the prescription and the choice stage. Importantly, these parameters can be identified using variation in the prescription stage alone. In particular, identification follows the same logic as the two-way fixed effects model used in our motivating evidence, following [Abowd et al. \(1999\)](#). Physician-specific prescribing propensities $\boldsymbol{\eta}$ are identified from variation in observed prescriptions for the same patient across different physicians. Similarly, the vector of patient opioid preference types $\boldsymbol{\Theta}$ is pinned down from variation in observed prescriptions written by the same physician for different patients. Note that this requires patients to switch exogenously between physicians. As in [Abowd et al. \(1999\)](#), parameters are identified only for physicians and patients who are connected through patient switching. We provide evidence of patients exogenously switching across physicians in section 3. The coefficients of the yearly fixed effects $\boldsymbol{\tau}$ are identified from within-patient variation in observed prescriptions across years, holding fixed both patient opioid preference types $\boldsymbol{\Theta}$ and physician prescribing propensities $\boldsymbol{\eta}$.

Next, conditional on the identification of Ξ_1 , we now discuss how data from the choice stage of our model identify the remaining parameters $\Xi_2 = \{\boldsymbol{\delta}, \sigma_\zeta, \boldsymbol{\phi}\}$. The physician mean utility parameters are physician-month fixed effects. Since our data contains observations for all patients and physicians in a given month, these parameters are identified from differences in observed physician choice frequencies, or market shares, within a month, holding fixed Z_{it} , the patient characteristics that affect the utility from an office visit. In other words, our micro data allows us to directly identify the physician mean utility parameters. Next, the demographic heterogeneity parameter vector $\boldsymbol{\phi}$ is identified from differences in the probability of seeking care across patients with different demographics Z_{it} , holding fixed physician mean utilities $\boldsymbol{\delta}$. Lastly, the dispersion of patient information types σ_ζ is identified from the time-persistent residual variation in patients' perceived prescription probabilities $\tilde{p}_{ijt}$ over time, conditional on patient opioid preference types $\boldsymbol{\Theta}$, physician prescribing propensities $\boldsymbol{\eta}$, and year fixed effects $\boldsymbol{\tau}$. Intuitively, patients choosing physicians differently from their optimal choice if they knew the true physician prescribing propensities systematically and persistently over time identifies the dispersion in patient information types, σ_ζ .

6 Estimation and Results

In this section, we describe how we estimate our structural model and present the estimation results.

6.1 Estimation

Since our model implies that in the second stage of prescribing, the physician does not consider her relative attractiveness to the patient or the information a patient has about her prescription type in the first stage, it is natural to estimate the choice stage and the prescription stage separately using a limited information maximum likelihood estimation procedure closely following our identification argument.

In particular, we estimate the model in three steps. In the first step, we focus on the prescription stage and estimate patients' opioid preference types Θ_i , physicians' prescribing types η_j , and yearly fixed effects τ that affect prescription decisions, using realized prescription decisions. In the second step, we take our prescription stage estimates as given and estimate the demand parameters in the choice stage. These include the mean utility associated with a physician δ_{jt} , the standard deviation of patients' information types σ_ζ , and the parameter vector ϕ , which governs time-varying patient characteristics that affect the utility of an office visit. Identification comes from variation in patients' physician choices. In the final stage, we decompose estimated patients' types Θ_i into guideline-concordant preferences (governed by β_1), non-concordant preferences reflected in observables (governed by β_2), and residual unobservable non-concordant preferences θ_i . We elaborate on each estimation step below.

Step I: Estimation of $\Xi_1 = \{\Theta, \eta, \tau\}$ from the prescription stage

We first note that the prescription stage depends only on $\Xi_1 = \{\Theta, \eta, \tau\}$. As shown in equation (14) in section 4, a physician's prescription probability is given by $\rho_{ijt} = \frac{\exp(\eta_j + \Theta_i + \tau_t)}{1 + \exp(\eta_j + \Theta_i + \tau_t)}$. We can reformulate this equation to the following two-way fixed effects logistic regression model:

$$\log\left(\frac{\rho_{ijt}}{1 - \rho_{ijt}}\right) = \eta_j + \Theta_i + \tau_t. \quad (18)$$

Note that this reduced-form equation derived from the prescription stage closely resembles our Abowd et al. (1999)-style specification in equation (4) for establishing patient sorting. It differs from equation (4) only due to the parametric type 1 extreme value assumption on the error term ε in physician's indirect utility.

Given its similarity to equation (4), equation (18) is subject to the same identification concerns. In particular, limited mobility bias might cause non-trivial bias when estimating the fixed effects (Andrews et al., 2012; Bonhomme et al., 2019). Moreover, logit models are known to suffer from incidental parameter bias when including fixed effects for each patient and physician. To address these issues, we follow Bonhomme et al. (2019) and Mourot (2025) and group physicians and patients in exactly the same manner as in section 3. We then estimate the grouped parameters using the prescription stage panel data via maximum likelihood. This step yields estimates of $\Xi_1 = \{\Theta, \eta, \tau\}$, which we denote by $\hat{\Xi}_1$ and take as given in step II.

Step II: Estimation of $\Xi_2 = \{\delta, \sigma_\zeta, \phi\}$ from the choice stage

Conditional on our estimates $\hat{\Xi}_1$ from the prescription stage, we estimate the demand side parameters δ , ϕ , and σ_ζ in a second step using maximum likelihood. Our approach is closely related to commonly used estimators in the health economics and industrial organization literature (Goolsbee and Petrin, 2004; Ho, 2006), but additionally leverages the panel dimension of our choice stage data.

Although in principle the choice stage of our model could be estimated via unconstrained maximum likelihood, doing so would require estimating thousands of δ parameters, which poses a substantial computational burden. Our data contain thousands of physicians observed over 84 months, making it impractical to estimate a separate parameter for each physician-month combination. We therefore follow the industrial organization literature and concentrate out the demand side mean utilities δ by imposing a market share constraint (Berry, 1994; Berry et al., 1995). We then maximize the constrained log-likelihood of the choice stage over $\Xi_2 = \{\sigma_\zeta, \phi\}$ only.

Conditional on our estimates from the prescription stage $\hat{\Xi}_1$, the constrained log-likelihood of the choice stage can be written as

$$\mathcal{L}_2 = \sum_{i=1}^N \log \int \prod_{t \in T_i} \prod_{j \in J_t} (s_{ijt}(\zeta; \Xi_2, \hat{\Xi}_1))^{y_{ijt}} f(\zeta; \sigma_\zeta) d\zeta \quad (19)$$

$$s.t. \quad s_{jt}(\Xi_2, \hat{\Xi}_1) = \mathcal{S}_{jt}, \quad (20)$$

where y_{ijt} is an indicator equal to 1 if patient i chooses physician j in month t , and 0 otherwise. $\mathcal{S}_{jt}$ represents the observed market share of physician j in month t in our data.

Because we use a two-step estimator, we need to adjust the standard errors of $\hat{\Xi}_2$ in the second stage to account for the first-step estimation. Since we use maximum likelihood for

both the first and the second step, it follows that under standard regularity conditions, the variance-covariance matrix for the second step, $\widehat{\mathbf{V}}_{\hat{\Xi}_2}$, is given by (Murphy and Topel, 2002):

$$\widehat{\mathbf{V}}_{\hat{\Xi}_2} = \widehat{\mathbf{V}}_2 + \widehat{\mathbf{V}}_2(\widehat{\mathbf{A}}\widehat{\mathbf{V}}_1\widehat{\mathbf{A}}')\widehat{\mathbf{V}}_2, \quad (21)$$

where $\widehat{\mathbf{V}}_1$ is the variance-covariance matrix from the first-step logit estimation, $\widehat{\mathbf{V}}_2$ denotes the variance-covariance matrix from the second-step likelihood estimation, and $\widehat{\mathbf{A}}$ is defined as

$$\widehat{\mathbf{A}} = \sum_i \left(\frac{\partial \mathcal{L}_{i2}}{\partial \hat{\Xi}_2} \right) \left(\frac{\partial \mathcal{L}_{i2}}{\partial \hat{\Xi}_1'} \right), \quad (22)$$

where $\mathcal{L}_{i2}$ denotes patient i 's individual contribution to the second-step likelihood. We provide the details of these derivations in Appendix G.

Step III: Separating Guideline-Concordant from Non-Concordant Patient Preferences

An important question for policy is whether prescription opioids are prescribed for medical reasons or instead reflect guideline-non-concordant preferences. To answer this question, we decompose our estimates of patients' preference type $\hat{\Theta}_i$ into three components: patients' guideline-concordant preferences (governed by β_1), observed non-concordant preferences (governed by β_2), and a residual unobservable non-concordant preference term θ_i . Note that among the observed non-concordant preferences, some may reflect medical consensus on managing chronic pain beyond the CDC guidelines. Specifically, physicians may also consider observable patient characteristics related to health status, including age, sex, the Charlson comorbidity index, counts of historical inpatient, outpatient, and emergency-department visits, and historical out-of-pocket spending. Because preferences supported by medical consensus may be fundamentally different from those related to opioid misuse, we further partition the observed guideline-non-concordant preferences into two components: conditions nonconcordant with guidelines but supported by medical consensus, and conditions concordant with neither guidelines nor medical consensus. We report the estimates for these two components separately.

We further adjust the corresponding standard errors of step III to account for the multi-step estimation procedure. Specifically, we adjust the standard errors analogous to equation (21) following Murphy and Topel (2002).

6.2 Results

Estimates of patients’ preference and information types, as well as physicians’ prescribing types, are reported in Table 5. We further decompose patients’ preference types in Table 6.

In Table 5, the estimated standard deviation of the patient information type, σ_ζ , indicates substantial dispersion in patients’ perceptions about physicians’ prescribing propensities. This suggests considerable heterogeneity both across patients and in how any given patient perceives different physicians. While some patients appear to be well informed, others substantially overestimate or underestimate their likelihood of obtaining prescription opioids from particular physicians during office visits.

Table 5 also shows that the patient preference type Θ is highly dispersed: some patients exhibit a strong preference for prescription opioids, while others exhibit a strong distaste for them. To gauge the magnitude, if a group 1 patient and a group 10 patient visit the middle group physician in the base year, the difference in their likelihood of receiving an opioid prescription is greater than 90%.

The decomposition of patient preference types in Table 6 shows that our estimates for patient preference types are strongly correlated with factors concordant with the guidelines or supported by medical consensus. For instance, we estimate that patients with sickle cell disease, a condition explicitly mentioned in CDC prescribing guidelines as warranting opioid prescriptions, exhibit a much higher opioid preference type than patients without the condition. Similarly, patients with higher comorbidity scores, as measured by the Charlson Comorbidity Index, and older patients tend to exhibit higher opioid preference. At the same time, our estimates are also strongly correlated with factors for which the CDC recommends caution in opioid prescribing. For example, previous opioid misuse or emergency room visits related to overdoses strongly and positively correlate with patients’ opioid preference.

Regarding patient heterogeneity in the utility of an office visit in Table 5, most patient characteristics do not significantly affect whether patients initiate an office visit. The exception is prior opioid misuse, which is associated with a significantly higher likelihood of visiting a physician.

We further estimate strong heterogeneity in prescribing propensities across physicians. For example, consider a physician A with the lowest prescribing type (group 2) and a physician B with the highest prescribing type (group 10). The likelihood of prescribing opioids for a patient with an average preference for opioids is 0.1% and 91%, respectively. These results suggest that the choice of physician meaningfully affects prescribing outcomes, regardless of patient opioid preference type.

Table 5: Estimated Model Parameters

Estimate (Std. Error)			Estimate (Std. Error)		
Patient Information Type			Patient Demographic Heterogeneity ϕ		
σ_{ζ}	5.133	(0.328)	Male	-0.028	(0.014)
Patient Preference Type Θ_i			Charlson Comorbidity Index	-0.003	(0.004)
			Opioid Misuse	0.013	(0.034)
			Substance Use Disorder	0.044	(0.019)
			Nonfatal Overdose	-0.012	(0.026)
Θ_1	-4.467	(0.590)	Age 20-24	0.203	(0.103)
Θ_2	-1.996	(0.171)	Age 25-29	0.290	(0.096)
Θ_3	0.300	(0.137)	Age 30-34	0.233	(0.092)
Θ_4	1.543	(0.133)	Age 35-39	0.231	(0.093)
Θ_5	2.006	(0.132)	Age 40-44	0.338	(0.093)
Θ_6	2.312	(0.133)	Age 45-49	0.267	(0.091)
Θ_7	2.760	(0.133)	Age 50-54	0.245	(0.090)
Θ_8	3.036	(0.135)	Age 55-59	0.215	(0.096)
Θ_9	3.649	(0.140)	Age 60-65	0.069	(0.099)
Θ_{10}	5.906	(0.197)			
Physician Prescribing Type η_j					
η_1	-8.678	(0.608)			
η_2	-6.102	(0.230)			
η_3	-3.915	(0.196)			
η_4	-3.573	(0.191)			
η_5	-3.351	(0.189)			
η_6	-3.227	(0.188)			
η_7	-3.092	(0.188)			
η_8	-2.909	(0.188)			
η_9	-2.604	(0.190)			
Constant	0.440	(0.227)			
Year FE	✓				
Data					
Patients (N)	11,909				
Physicians (J)	8,059				
Months (T)	84				
Total Observations	126,357				

Notes: This table reports estimated model parameters from the structural model. Standard errors are adjusted for two-step estimation and reported in parentheses. The omitted age group is below the age of 20.

Table 6: Projecting estimated patient opioid preference onto guideline-concordant and non-concordant parts

	Estimate	(Std. Error)
Conditions Concordant with Guidelines X_{1i}		
Cancer Pain	0.162	(0.107)
Sickle Cell Disease	1.183	(0.346)
Conditions Nonconcordant with Guidelines but Supported by Medical Consensus X_{2i}		
Male	0.186	(0.046)
Outpatient Spend (avg.)	-0.004	(0.000)
Inpatient Spend (avg.)	0.187	(0.066)
Charlson CCI (avg.)	0.087	(0.020)
OOP Visits (avg.)	0.000	(0.000)
ER Visits (avg.)	-0.015	(0.019)
OOP Drugs (avg.)	0.274	(0.062)
Age 20-24	0.005	(0.270)
Age 25-29	0.078	(0.255)
Age 30-34	0.356	(0.256)
Age 35-39	0.409	(0.257)
Age 40-44	0.599	(0.276)
Age 45-49	1.525	(0.476)
Age 50-54	1.539	(0.498)
Age 55-59	1.731	(0.497)
Age 60-65	2.533	(0.494)
Conditions Concordant with Neither Guidelines nor Medical Consensus X_{2i}		
ER Visit Overdose	0.750	(0.473)
Renal/Hepatic Insufficiency	-0.240	(0.150)
Mental Health Condition	0.059	(0.049)
Opioid Misuse	1.572	(0.133)
Substance Use Disorder	0.308	(0.071)
Nonfatal Overdose	0.173	(0.127)
Safety Profession	0.107	(0.082)
Constant	-1.830	(0.405)
Data		
Patients (N)		11,909

Notes: OLS estimates from regressions of Θ_i on covariate sets X_{1i} and X_{2i} . Standard errors are adjusted for two-step estimation and reported in parentheses. The omitted age group is below the age of 20.

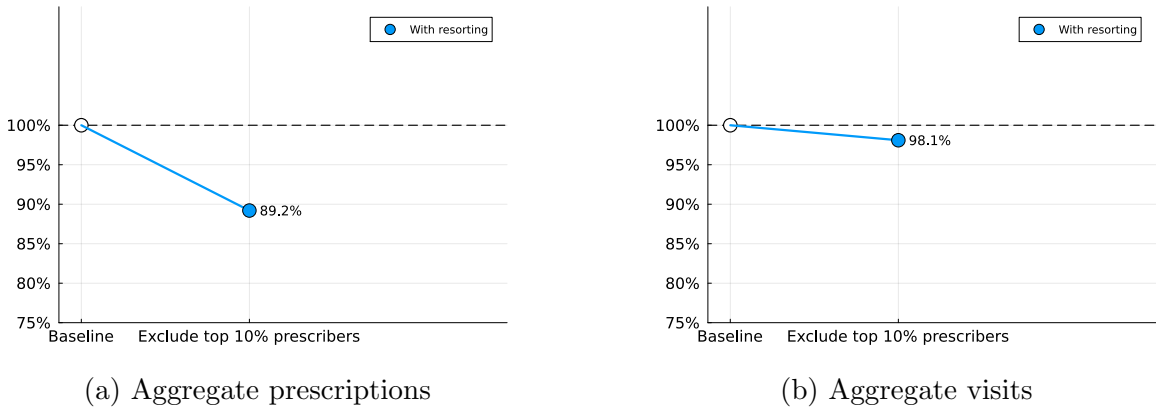
7 Counterfactual Simulations

In this section, we demonstrate how patient sorting can influence the effectiveness of top provider-targeted policies. Moreover, we show that an alternative provider-targeted policy, which also eliminates prescriptions driven by patient preferences exceeding medical consensus or CDC guidelines, achieves a larger and more targeted reduction in aggregate prescriptions. We highlight the underlying economic mechanisms below and provide technical details on our implementation in Appendix H.

7.1 Excluding Top Prescribers

We start by simulating the *exclusion* of the top 10% of physicians by risk-adjusted prescription rate across all markets.¹⁵ This policy prevents patients from choosing any of the excluded physicians. Our counterfactual exclusion policy closely mirrors real-world policies such as the Drug Enforcement Administration’s prescription license revocation policy for opioid prescribing physicians, but is more extreme in the percentage of excluded physicians to illustrate the economic mechanisms and effects of these policies. We simulate the counterfactual choices for all patients who see an excluded physician in the baseline. Patients who chose an excluded physician in the baseline may either choose to see any of the remaining, non-excluded physicians or the outside option of not seeing any physician in that month. We fix the choices of patients who did not see an excluded physician.

Figure 8: Excluding the top 10% of physicians by risk-adjusted prescription rate



Notes: This figure plots the aggregate prescriptions and visits after excluding the top 10% of physicians by risk-adjusted prescription rate. The numbers are expressed as percentages relative to the baseline.

The left panel of Figure 8 compares the counterfactual aggregate prescriptions in percent-

¹⁵We define the risk-adjusted prescription rate based on Equation (2).

ages relative to the baseline. Aggregate opioid prescriptions are a key metric for policymakers because opioid prescribing in the primary market is closely associated with drug overdoses and mortality, with the majority of secondary-market opioid supply originating from the primary market (Schnell, 2025). Figure 8 shows that our policy reduces aggregate prescriptions by only around 11%, while aggregate visits drop by around 2%. In light of the evidence presented in section 3 that the distribution of physicians’ prescription intensity is highly skewed, with the top 10% of physicians in terms of raw prescription rates accounting for 30% of aggregate prescriptions, this muted response is surprising. Given the skewed distribution, everything else equal, excluding high-prescribing physicians should reduce opioid prescription rates more dramatically.¹⁶

However, this argument ignores the demand side response to top physician-targeted policies. To illustrate how the exclusion policy leads to patient choice reoptimization, we plot the average choice probability for each physician prescribing propensity type group η across each patient benefit type group Θ . Figure 9 shows that patients with high opioid benefit type Θ are disproportionally more likely to choose physicians who prescribe opioids more frequently, regardless of patients’ benefit type. In other words, when we limit patients’ choice sets to a subset of physicians, affected patients will reoptimize by choosing the physician who, *ceteris paribus*, most closely aligns with their preferences for prescription opioids. However, Figure 9 also demonstrates that resorting is not perfectly aligned with patients automatically choosing the highest prescribers. Our demand system flexibly captures differences in the attractiveness of physicians, such as quality of care, friendliness of front desk staff, etc., via the mean utility terms δ . Our estimates imply that these differences matter. In particular, we show that although patients with positive benefit type Θ , groups Θ_3 to Θ_{10} , exhibit a strong preference for prescription opioids, patients are slightly more likely to trade off a higher probability for obtaining a prescription for better quality of care. Similarly, patients who dislike prescription opioids, groups Θ_1 and Θ_2 , accept higher prescription probabilities for better quality of care. Nevertheless, Figure 9 clearly demonstrates that patients do not randomly choose physicians in response to exclusion policies. Instead, patients resort to physicians.

Next, notice that aggregate visits decline under the exclusion policy as illustrated in the right panel of Figure 8. This implies that a fraction of the observed reduction in the aggregate

¹⁶Given that the top 10% of physicians in terms of raw prescription rates accounting for 30% of aggregate prescriptions, if we naively assume that affected patients would randomly switch to the remaining physicians, a simple back-of-the-envelope calculation—without the structural model—would suggest that this counterfactual reduces aggregate prescriptions by about 22 percentage points, even without considering the possibility of patient forgo care. However, our structural model estimates show that this policy reduces aggregate prescriptions by only 11 percentage points.

Figure 9: Average choice probabilities for each patient group (Θ group) by physician group (η group).

η_{10}	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%
η_9	0.099%	0.097%	0.105%	0.126%	0.129%	0.134%	0.143%	0.147%	0.145%	0.147%
η_8	0.141%	0.137%	0.146%	0.166%	0.171%	0.174%	0.183%	0.184%	0.184%	0.180%
η_7	0.164%	0.159%	0.167%	0.185%	0.188%	0.190%	0.198%	0.197%	0.193%	0.190%
η_6	0.168%	0.164%	0.173%	0.188%	0.192%	0.194%	0.198%	0.198%	0.195%	0.197%
η_5	0.209%	0.204%	0.212%	0.228%	0.229%	0.229%	0.234%	0.234%	0.230%	0.226%
η_4	0.180%	0.177%	0.180%	0.190%	0.192%	0.194%	0.200%	0.199%	0.197%	0.201%
η_3	0.166%	0.163%	0.162%	0.169%	0.169%	0.170%	0.176%	0.176%	0.172%	0.175%
η_2	0.212%	0.212%	0.186%	0.153%	0.140%	0.134%	0.130%	0.124%	0.114%	0.121%
η_1	0.182%	0.188%	0.155%	0.107%	0.089%	0.080%	0.071%	0.065%	0.058%	0.059%
	Θ_1	Θ_2	Θ_3	Θ_4	Θ_5	Θ_6	Θ_7	Θ_8	Θ_9	Θ_{10}

Notes: Average choice probability of patients conditional on choosing a physician in the market. Patient and physician groups are in ascending order.

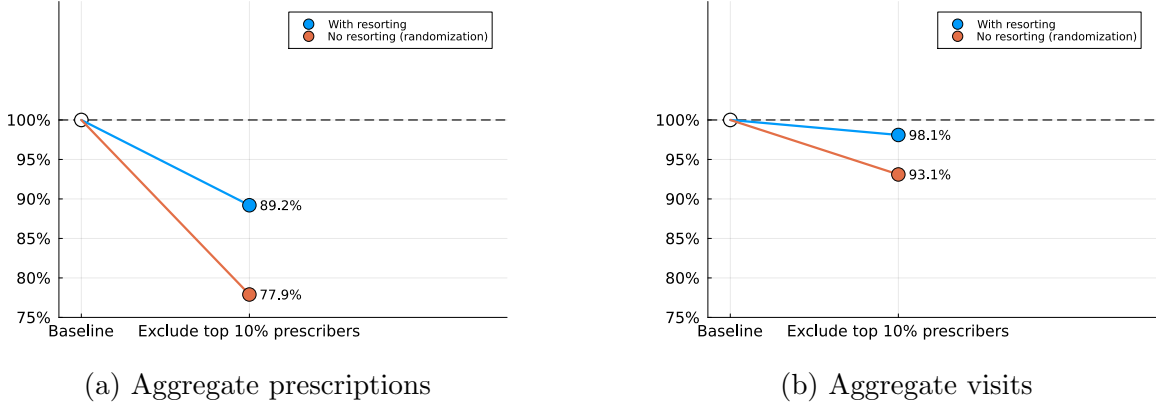
prescription rate is driven by patients forgoing care. We further decompose the reduction in aggregate prescriptions into the contribution of affected patient switching to lower prescribers (conditional on market participation) and the contribution of patients forgoing care.¹⁷ Our results imply that 74% of the predicted reduction in aggregate prescriptions is attributable to patients being forced to switch to physicians with lower prescription propensity, while the remaining 26% is caused by patients forgoing care.

7.2 Excluding Top Prescribers & No Resorting

A surprising outcome of our first counterfactual policy is that the exclusion policy's effect is attenuated by affected patients resorting to remaining physicians who are, everything else equal, most likely to prescribe opioids. In this counterfactual, we quantify the importance of the patient resorting effect by explicitly prohibiting resorting, i.e., preventing affected patients from choosing their physicians. We simulate a counterfactual in which we allow patients to only choose between being randomly assigned to a physician and not seeking care in a given month. As before, we hold fixed all choices of unaffected patients.

¹⁷To implement this decomposition, we simulate the reduction in prescription by keeping every patient in the market and requiring each to choose an inside option. We then compare this reduction to the overall

Figure 10: Excluding the top 10% of physicians and banning patient resorting



Notes: This figure plots the aggregate prescriptions and visits after excluding the top 10% of physicians by risk-adjusted prescription rate and randomly assigning patients to the remaining physicians. The numbers are expressed as percentages relative to the baseline.

Figure 10 illustrates that, without allowing patients to reoptimize their choices of physicians, the exclusion policy studied in section 7.1 would reduce the opioid prescription rate by an additional 11 percentage points. When policymakers naïvely ignore patient resorting, expected policy effects may be substantially overstated.

If a policy focuses solely on aggregate prescriptions, top provider-targeted policies combined with restrictions on patient choices appear effective. However, while banning resorting can substantially reduce aggregate prescribing, this policy comes at a cost: patients forgoing care. Note that compared to section 7.1, the additional reduction in aggregate prescriptions originates from two channels. First, eliminating resorting prevents patients who remain in the market from making optimal choices about their physician. This eliminates resorting to physicians whose prescribing propensities are, *ceteris paribus*, close to those of their original physicians who got excluded. Second, prohibiting resorting also induces some patients to forgo care altogether, as random assignment makes the outside option relatively more attractive. The right panel of Figure 10 shows that aggregate physician visits drop by an additional 5 percentage points when resorting is prohibited. Preventing affected patients from choosing their physician optimally based on their preferences leads many to forgo care. Mechanically, this happens because we estimate substantial heterogeneity in benefit types Θ and prescribing propensity types η . Random assignment makes it very likely that a patient will be matched with a physician whose prescribing type does not align with the patient's preference.

We further decompose the additional 11 percentage points reduction in aggregate pre-reduction to isolate the portion attributable to patients forgoing care.

scriptions into the contribution of banning resorting (conditional on market participation) and the contribution of patients forgoing care. We find that 8% of the reduction under randomization is due to the elimination of patient resorting, while the remaining 92% is driven by increased patient exit. This means that even patients who would qualify for opioid treatment under standard clinical guidelines or medical consensus may be prevented from obtaining appropriate care. This constitutes a clearly undesirable policy outcome.

In our next counterfactual, we explicitly address the trade-off between limiting guideline-non-concordant preference-based sorting and the risk of denying access to clinically appropriate opioid prescriptions.

7.3 Breaking Bad Opioid Sorting

We now consider a policy that still excludes the top prescribers but, for affected patients, preserves opioid treatment when preferences are supported by guidelines or medical consensus while eliminating prescriptions driven by preferences concordant with neither. To do so, we categorize sorting driven by patient preferences concordant with CDC guidelines or medical consensus as “good sorting” and the remainder as “bad sorting.” In our counterfactual simulation, we enforce that physicians prescribe only considering preferences concordant with CDC guidelines or medical consensus. Essentially, this policy breaks bad sorting while preserving good sorting. As before, we hold fixed all choices of unaffected patients.

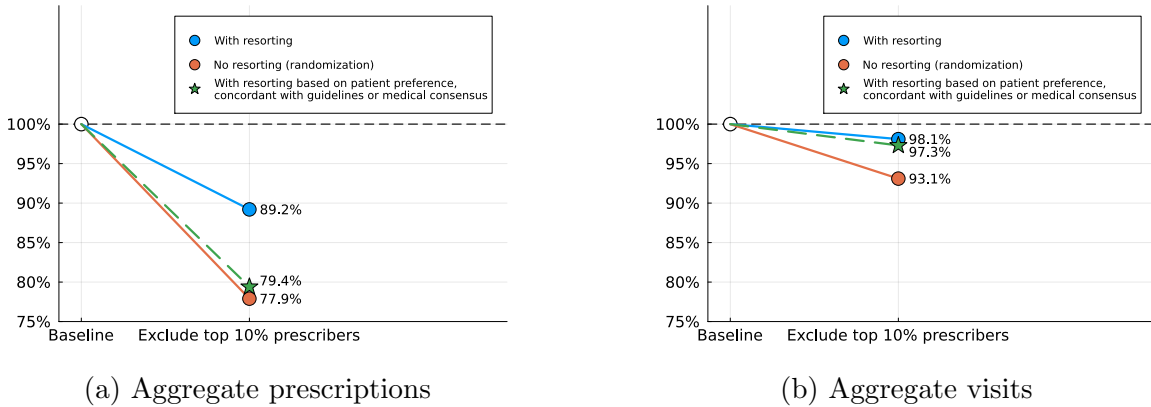
Since we only observe part of the medical consensus, specifically, prescribing based on patient severity factors such as age and comorbidity, we cannot determine with certainty how much of the residual unobserved preference θ_i reflects preferences supported by medical consensus. For example, physicians may all agree to practice compassionate care for patients experiencing severe pain, but pain levels are unobserved in our data. To address this, we bound the policy’s effect with two extreme assumptions. At one extreme, we assume that all unobserved patient preferences are inconsistent with medical consensus, implying that we observe all relevant medical consensus and that all sorting based on unobserved preferences constitutes bad sorting. At the other extreme, we assume that all unobserved patient preferences support opioid prescribing based on medical consensus and thus constitute good sorting that should be preserved.¹⁸ We also consider intermediate cases in which a portion of θ_i is attributed to bad sorting.

Specifically, for each patient affected by the exclusion policy, we require physicians to prescribe based on the following rules:

¹⁸We note that attributing all sorting based on patient unobserved preference to good sorting is unrealistic. Recent evidence suggests that prescribing based on medical consensus is responsible for a significant fraction of U.S. drug deaths (Evans and Lieber, 2026).

1. A physician will consider the full benefit type Θ if the patient suffers from sickle cell disease, cancer-related pain, or is in end-of-life care or palliative care. In other words, we do not interfere with opioid prescribing for chronic pain associated with conditions that are explicitly identified as medically warranted in the CDC guidelines.
2. In all other cases, if a patient's preference related to bad sorting is positive, that is, the sum of observed preferences concordant with neither the guidelines nor medical consensus plus a portion of θ_i (depending on assumptions about the share unsupported by medical consensus), the physician should ignore this preference. If the patient's preference related to bad sorting is negative, we do not intervene, and the physician should take this preference into account.¹⁹

Figure 11: Excluding the top 10% of physicians and breaking sorting based on preferences concordant with neither guidelines nor medical consensus



Notes: This figure plots the aggregate prescriptions and visits after excluding the top 10% of physicians by risk-adjusted prescription rate and eliminating sorting based on preferences concordant with neither guidelines nor medical consensus, under the assumption that all unobserved patient preferences are unsupported by medical consensus and thus eliminated. The numbers are expressed as percentages relative to the baseline.

Figure 11 shows the results under the assumption that none of the residual preferences is supported by medical consensus. Eliminating bad sorting results in almost the same reduction in opioid prescriptions as the randomization policy. Turning to patient visits, however, we find a stark contrast: unlike randomization, enforcing more stringent prescribing guidelines leads to only a moderate decline in office visits, comparable to the counterfactual that allows resorting. In other words, our counterfactual policy eliminates bad resorting almost as effectively as randomization while largely keeping patients in care.

¹⁹If a patient has any conditions mentioned in the CDC guideline (CDC2016) that warrant caution, physicians must not prescribe opioids for pain based on this preference.

Table 7 shows the results from varying the assumed share of θ_i supported by medical consensus. The 100% column corresponds to the extreme case in Figure 11 where none of the unobserved preferences are supported by medical consensus. The 0% column displays results for the other extreme, where all unobserved preferences are supported by medical consensus and physicians are allowed to consider them when prescribing. Even in this most conservative case, aggregate prescriptions drop by 13% while a comparable number of patients remain in care.

Table 7: Changes in aggregate prescriptions and visits under different assumptions

Share of residuals unsupported by medical consensus	100%	75%	50%	25%	0%
$\Delta_{\text{Prescriptions}}$ (pp)	-20.6	-19.6	-17.7	-15.1	-12.2
Δ_{Visits} (pp)	-2.7	-2.5	-2.3	-2.1	-2.0

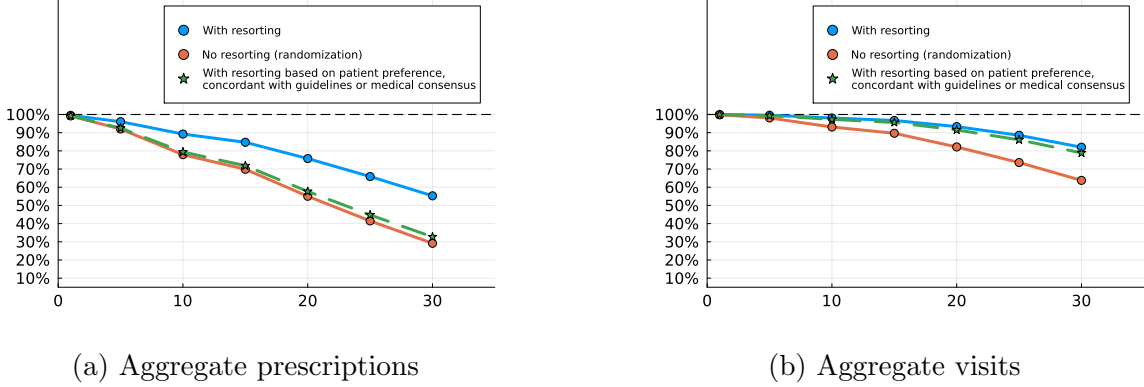
Notes: Each column reports changes in aggregate prescriptions and visits under a scenario in which a given share of θ_i is assumed to be unsupported by medical consensus.

Figure 12 further shows that these results do not depend on the choice of cutoff value used to exclude physicians. Such results further suggest that policy could eliminate discretionary prescribing for all physicians, rather than targeting only top prescribers. Our final counterfactual explores this possibility by applying the prescription rules to all physicians, eliminating bad sorting for every patient. We find that a policy eliminating discretionary prescribing for all physicians, without excluding any physicians from the market, is approximately as effective as excluding the top 37% of physicians by risk-adjusted prescription rate, while maintaining access to care for patients with preferences consistent with guidelines and medical consensus.

7.4 Patient Information on Physician Type Facilitates Resorting

In Section 7.2, we demonstrated how resorting may attenuate the effectiveness of exclusion policies. We now highlight the role of patient information about physician prescribing propensities in exacerbating resorting and further attenuating the policy’s effect. Intuitively, when patients have perfectly accurate information about physician types, they can sort more effectively into physicians aligned with their preferences. In the baseline, some high-type patients visit low-type physicians because they receive noisy signals about those physicians’ types. With accurate information, sorting becomes more precise: high-type patients are systematically more likely to match with high-type physicians. As a result, we would expect an

Figure 12: Counterfactual results by physician exclusion cutoff

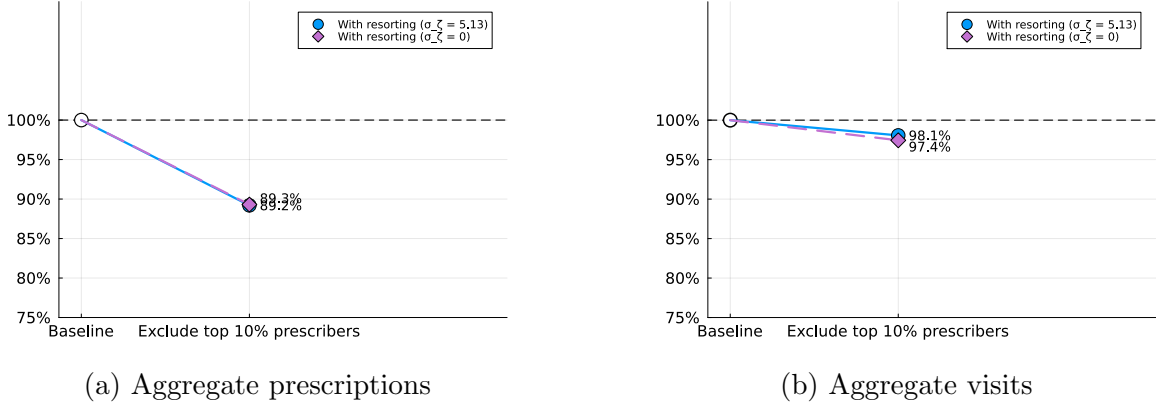


Notes: Prescription rates and visits across all three counterfactuals as a function of exclusion percentile and setting the share of θ_i unsupported by medical consensus to 100%. The numbers are expressed as percentages relative to the baseline.

even smaller reduction in opioid prescriptions following the exclusion policy with resorting. On the other hand, better information may lead some patients to leave the market as it reveals that no remaining physician is sufficiently attractive to justify a visit.

Figure 13 compares the effects of the exclusion policy described in section 7.1 between two information environments: the baseline environment, where patients have noisy information about physicians' prescribing types as estimated, and a counterfactual information environment where patients have full information about physicians' prescribing types. Despite a decrease in total visits, the reduction in aggregate prescription is attenuated in the counterfactual information environment when resorting becomes more precise. This finding highlights a trade-off in empowering patients with information. Conventional wisdom holds that patient information is a powerful tool for resolving choice frictions and disciplining physicians. Here, we highlight that information about physicians' prescribing types may unintentionally facilitate more precise bad sorting.

Figure 13



Notes: These figures plot the aggregate prescriptions and visits relative to the baseline after excluding the top 10% of physicians by risk-adjusted prescription rate. The numbers are expressed as percentages relative to the baseline.

8 Conclusion

In this paper, we provide novel evidence on how consumers sort into experts based on aligned preferences. We demonstrate that patients with chronic pain strongly sort into physicians based on their preferences for prescription opioids. We then estimate a structural model of patient sorting and physician prescribing to understand the policy implications of patient sorting.

Through counterfactual analyses, we quantify how sorting undermines policies that target high-prescribing physicians, substantially attenuating their effectiveness. Moreover, nearly all of this attenuation of top physician-targeted policies arises from resorting driven by patient preferences concordant with neither clinical guidelines nor medical consensus. We further show that relatively straightforward policies that eliminate physician discretion can fully eliminate undesirable patient sorting while still preserving patients' necessary clinical needs.

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A Charlson Comorbidity Index (CCI)

We use weights in [Schneeweiss et al. \(2003\)](#) for each of the 17 conditions suggested for the Charlson/Deyo variant ([Deyo, 1992](#)). The 17 conditions are:

Comorbid Condition	ICD10 Diagnosis Codes	Weights
Myocardial Infarction	I21, I22, I25.2	1
Congestive Heart Failure	I09.9, I11.0, I13.0, I13.2, I25.5, I42.0, I42.5–I42.9, I43, I50, P29.0	1
Peripheral Vascular Disease	I70, I71, I73.1, I73.8, I73.9, I77.1, I79.0, I79.2, K55.1, K55.8, K55.9, Z95.8, Z95.9	1
Cerebrovascular Disease	G45, G46, H34.0, I60–I69	1
Dementia	F00–F03, F05.1, G30, G31.1	1
Chronic Pulmonary Disease	I27.8, I27.9, J40–J47, J60–J67, J68.4, J70.1, J70.3	1
Connective Tissue Disease	M05, M06, M31.5, M32–M34, M35.1, M35.3, M36.0	1
Peptic Ulcer Disease	K25–K28	1
Mild Liver Disease	B18, K70.0–K70.3, K70.9, K71.3–K71.5, K71.7, K73, K74, K76.0, K76.2–K76.4, K76.8, K76.9, Z94.4	1
Diabetes w/o Complications	E10.0–E10.1, E10.6, E10.8–E10.9, E11.0–E11.1, E11.6, E11.8–E11.9, E12.0–E12.1, E12.6, E12.8–E12.9, E13.0–E13.1, E13.6, E13.8–E13.9, E14.0–E14.1, E14.6, E14.8–E14.9	1
Diabetes w/ Chronic Complications	E10.2–E10.5, E10.7, E11.2–E11.5, E11.7, E12.2–E12.5, E12.7, E13.2–E13.5, E13.7, E14.2–E14.5, E14.7	2
Paraplegia and Hemiplegia	G04.1, G11.4, G80.1, G80.2, G81, G82, G83.0–G83.4, G83.9	2
Renal Disease	I12.0, I13.1, N03.2–N03.7, N05.2–N05.7, N18, N19, N25.0, Z49.0–Z49.2, Z94.0, Z99.2	2
Cancer	C00–C26, C30–C34, C37–C41, C43, C45–C58, C60–C76, C81–C85, C88, C90–C97	2
Moderate or Severe Liver Disease	I85.0, I85.9, I86.4, I98.2, K70.4, K71.1, K72.1, K72.9, K76.5–K76.7	3
Metastatic Carcinoma	C77–C80	6
HIV/AIDS	B20–B22, B24	6

Table 8: Charlson Comorbidity Index: ICD10 Diagnoses and Weights

Patients in the following age groups receive additional points for their index:

Age Groups	Weights
<50 years	0
50–59 years	1
60–69 years	2
70–79 years	3
≥80 years	4

Table 9: Charlson Comorbidity Index: Age Groups and Weights

B CDC Clinical Practice Guideline for Prescribing Opioids for Pain

CDC2016 provides recommendations for clinicians providing pain care, including those prescribing opioids, for outpatients aged ≥ 18 years. For section 7.3, we follow the guidelines and identify the medical conditions that are concordant with CDC guidelines (Table 10) and neither concordant with CDC guidelines nor medical consensus (Table 11) opioid prescriptions. In our data, we do not observe any patients receiving the palliative/ end-of-life care diagnosis.

Medical Diagnosis	ICD10 Diagnosis Codes
sickle cell disease	D57
cancer-related pain	G89.3
palliative / end-of-life care	Z51.5

Table 10: Medical conditions that are concordant with CDC guidelines

Medical Diagnosis	ICD10 Diagnosis Codes
sleep-disordered breathing	G47.3
renal or hepatic insufficiency	N17, K72
mental health condition (anxiety, depression and post-traumatic stress disorder)	F32, F41, F43
opioid abuse	F11, T40
substance use disorder	F10, F12-F19
non-fatal overdose	T36 - T39, T41 - T50

Table 11: Medical conditions that are neither concordant with CDC guidelines nor medical consensus

C Constructing expected out-of-pocket prescription opioid costs

In our data, out-of-pocket opioid costs are observed only for chronic pain patients who fill an opioid prescription. We construct a measure of expected out-of-pocket costs by computing the average opioid OOP cost at the plan-year-month level among observed users.

$$O\bar{O}PD_{it} = \frac{1}{N_{plan(i)t}} \sum_{i' \in plan(i)t} OOPD_{i't}$$

$N_{plan(i)t}$ is the number of patients in the same plan as patient i in year-month t that had an opioid prescription. $O\bar{O}PD_{it}$ is the average OOPD for all patient i' that are in the same insurance plan as patient i in year-month t .

D Grouping Algorithm

To group both patients and providers, we first run two regressions to construct the average risk-adjusted prescription rate across patients for providers, and the average risk-adjusted prescription rate across providers for patients. In particular, we run

$$prescribe_{ijt} - X_{it}\beta = \tilde{\eta}_j + \epsilon_{ijt}, \quad (23)$$

and

$$prescribe_{ijt} - X_{it}\beta = \tilde{\theta}_i + \epsilon_{ijt}, \quad (24)$$

where X_{it} is the patient characteristics defined in section 3. Whenever available, we use the time-varying versions of these variables, including the Charlson comorbidity index, counts of historical inpatient, outpatient, emergency-department visits, and emergency-department visits related to overdose, historical out-of-pocket spending, and insurance cost sharing. We then use our estimates for $\tilde{\eta}_j$ and $\tilde{\theta}_i$ to group patients and providers using the k-mean clustering algorithm (Bonhomme et al., 2019) as in Mourot (2025). In the second stage, we can then consistently estimate the two-way fixed effects model in equation (4) using our groups from the first stage.

E Figures

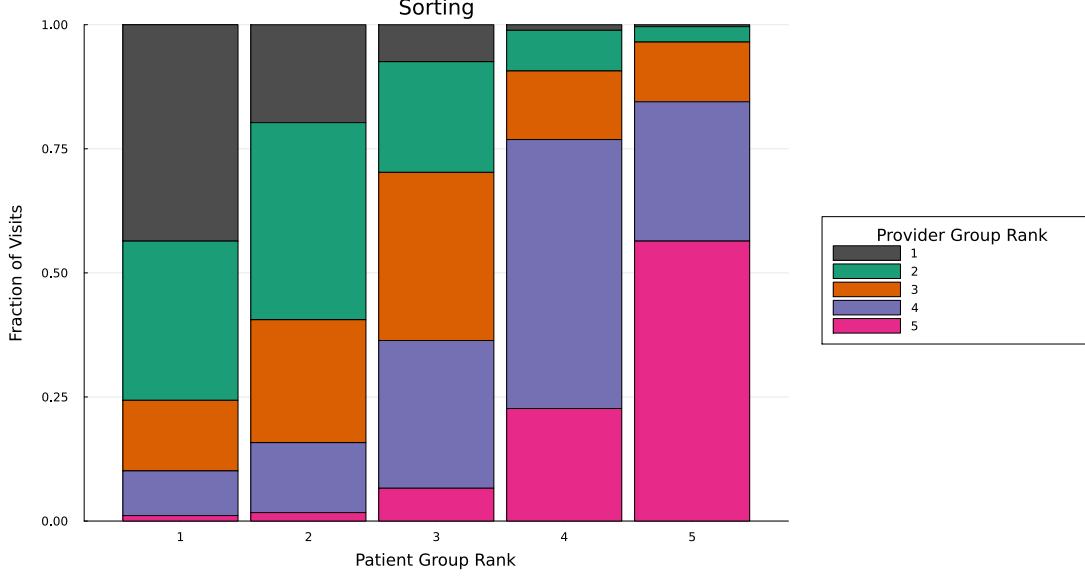


Figure 14: Sorting

F Constructing a measure of opioid misuse

Following the literature, we classify an outpatient visit as involving opioid misuse if the patient’s daily Morphine Equivalent Dosage (MED) exceeds 120 mg, given the documented correlation between such prescription levels and opioid abuse (Meara et al., 2016; Staiger et al., 2022; Finkelstein et al., 2025). In this section, we describe how we calculate the MED for an outpatient visit.

For a visit t , we observe all prescription drugs d , their active opioid ingredients o , and the respective dosages in milligrams (mg).²⁰ Following the CDC guidelines (CDC2016), we consider commonly prescribed opioid ingredients in pain management: codeine, fentanyl, hydrocodone, hydromorphone, methadone, morphine, oxycodone, oxymorphone, tapentadol, and tramadol. For each prescription drug d , we define its morphine equivalent dosage (MED) as the sum of all opioids’ dosage multiplied by the conversion factor:

$$MED_d = \sum_{o \in N_d^{opioid}} dosage_o \times conversion_o$$

N_d^{opioid} denotes all the opioid ingredients found in drug d . $conversion_o$ is the conversion factor used by the CDC to standardize the comparisons between opioids relative to morphine and is found in (CDC2016). Hence, patient i ’s daily MED in outpatient visit t is defined as the sum of all drugs’ total daily MED:

²⁰Dosages in micrograms (mcg) are divided by 1,000 to convert to milligrams (mg).

$$MED_{it} = \sum_{d \in N_{it}^{drug}} \frac{MED_d \times METQTY_{dit}}{DAYSUPP_{dit}}$$

N_{it}^{drug} denotes all the drugs prescribed to patient i in outpatient visit t . $METQTY_{dit}$ is the metric quantity of drug d for patient i in outpatient visit t (i.e., number of pills prescribed). $DAYSUPP_{dit}$ is drug d 's days of supply for patient i in outpatient visit t . Hence, an outpatient visit t is defined as an opioid misuse if MED_{it} exceeds 120 mg.

G Gradient for likelihood equation 19

In this section, we describe how we derive the gradient for the likelihood equation 19. We use these derivations to calculate the Murphy-Topel adjusted standard errors in equations 21 and 22.

We can now derive the following partial derivatives. For each element m of the ϕ vector and with some abuse of notation (integral vs sum), we have

$$\frac{\partial s_{jt}}{\partial \phi^{\{m\}}} = \sum_i \int_{-\infty}^{+\infty} \frac{\partial s_{ijt}}{\partial \phi^{\{m\}}} dF(\zeta) = \sum_i \int_{-\infty}^{+\infty} z_i^{\{m\}} s_{ijt} s_{i0t} dF(\zeta). \quad (25)$$

$$\frac{\partial s_{jt}}{\partial \sigma_\zeta} = \sum_i \frac{\partial}{\partial \sigma_\zeta} \int_{-\infty}^{+\infty} s_{ijt} dF(\zeta) \quad (26)$$

Let g denote a specific physician group. For each element g of the η vector, we consider:

- if physician $j \in g$

$$\begin{aligned} \frac{\partial s_{jt}}{\partial \eta_j^{\{g\}}} &= \sum_i \int_{-\infty}^{+\infty} \frac{\partial s_{ijt}}{\partial \eta_j^{\{g\}}} dF(\zeta) \\ &= \sum_i \int_{-\infty}^{+\infty} s_{ijt} (\Theta_i + \alpha X_{it}) \left[\tilde{\rho}_{ijt} (1 - \tilde{\rho}_{ijt}) - \sum_{l \in g} \tilde{\rho}_{ilt} (1 - \tilde{\rho}_{ilt}) s_{ilt} \right] dF(\zeta). \end{aligned} \quad (27)$$

- if physician $j \notin g$

$$\begin{aligned} \frac{\partial s_{jt}}{\partial \eta_j^{\{g\}}} &= \sum_i \int_{-\infty}^{+\infty} \frac{\partial s_{ijt}}{\partial \eta_j^{\{g\}}} dF(\zeta) \\ &= - \sum_i \int_{-\infty}^{+\infty} s_{ijt} (\Theta_i + \alpha X_{it}) \left[\sum_{l \in g} \tilde{\rho}_{ilt} (1 - \tilde{\rho}_{ilt}) s_{ilt} \right] dF(\zeta). \end{aligned} \quad (28)$$

Similarly, for the constant term, we have

$$\begin{aligned}\frac{\partial s_{jt}}{\partial cons} &= \sum_i \int_{-\infty}^{+\infty} \frac{\partial s_{ijt}}{\partial cons} dF(\zeta) \\ &= \sum_i \int_{-\infty}^{+\infty} s_{ijt}(\Theta_i + \alpha X_{it}) \left[\tilde{\rho}_{ijt}(1 - \tilde{\rho}_{ijt}) - \sum_l \tilde{\rho}_{ilt}(1 - \tilde{\rho}_{ilt})s_{ilt} \right] dF(\zeta).\end{aligned}\quad (29)$$

Let k denote a specific patient group. For each element k of the Θ vector, we have:

$$\frac{\partial s_{jt}}{\partial \Theta_i^{\{k\}}} = \sum_i \int_{-\infty}^{+\infty} \frac{\partial s_{ijt}}{\partial \Theta_i^{\{k\}}} dF(\zeta) \quad (30)$$

with:

$$\frac{\partial s_{ijt}}{\partial \Theta_i^{\{k\}}} = \begin{cases} s_{ijt} [(\Theta_i + X_{it}\alpha) (\tilde{\rho}_{ijt}(1 - \tilde{\rho}_{ijt}) - \sum_l s_{ilt}\tilde{\rho}_{ilt}(1 - \tilde{\rho}_{ilt})) + \tilde{\rho}_{ijt} - \sum_l s_{ilt}\tilde{\rho}_{ilt}] & \text{if } i \in k \\ 0 & \text{if } i \notin k \end{cases}$$

For each element m of α , we have

$$\begin{aligned}\frac{\partial s_{jt}}{\partial \alpha^{\{m\}}} &= \sum_i \int_{-\infty}^{+\infty} \frac{\partial s_{ijt}}{\partial \alpha^{\{m\}}} dF(\zeta) \\ &= \sum_i \int_{-\infty}^{+\infty} s_{ijt} X_{it}^{\{m\}} \left[(\Theta_i + X_{it}\alpha) \left(\tilde{\rho}_{ijt}(1 - \tilde{\rho}_{ijt}) - \sum_l s_{ilt}\tilde{\rho}_{ilt}(1 - \tilde{\rho}_{ilt}) \right) \right. \\ &\quad \left. + \tilde{\rho}_{ijt} - \sum_l s_{ilt}\tilde{\rho}_{ilt} \right] dF(\zeta).\end{aligned}\quad (31)$$

For the partial derivatives with respect to δ , we have

$$\frac{\partial s_{jt}}{\partial \delta_{jt}} = \sum_i \int_{-\infty}^{+\infty} \frac{\partial s_{ijt}}{\partial \delta_{jt}} dF(\zeta) = \sum_i \int_{-\infty}^{+\infty} s_{ijt}(1 - s_{ijt}) dF(\zeta), \quad (32)$$

and

$$\frac{\partial s_{jt}}{\partial \delta_{lt}} = \sum_i \int_{-\infty}^{+\infty} \frac{\partial s_{ijt}}{\partial \delta_{lt}} dF(\zeta) = \sum_i \int_{-\infty}^{+\infty} -s_{ijt}s_{ilt} dF(\zeta). \quad (33)$$

The fixed point imposes that the predicted shares are equal to the observed shares. That is, $0 = s_{jt}(\Xi) - \mathcal{S}_{jt}$. By the implicit function theorem, we have that

$$\left(\frac{d\delta_t}{d\sigma_\zeta} \quad \frac{d\delta_t}{d\phi} \quad \frac{d\delta_t}{d\eta} \quad \frac{d\delta_t}{dcons} \quad \frac{d\delta_t}{d\Theta} \quad \frac{d\delta_t}{d\alpha} \right) = - \left(\frac{\partial \mathbf{s}_t}{\partial \delta_t} \right)^{-1} \left[\frac{\partial \mathbf{s}_t}{\partial \sigma_\zeta} \quad \frac{\partial \mathbf{s}_t}{\partial \phi} \quad \frac{\partial \mathbf{s}_t}{\partial \eta} \quad \frac{\partial \mathbf{s}_t}{\partial cons} \quad \frac{\partial \mathbf{s}_t}{\partial \Theta} \quad \frac{\partial \mathbf{s}_t}{\partial \alpha} \right].$$

Now, we can use these to form the components of the Murphy-Topel adjusted standard errors, which consist of the gradients of the loglikelihood with respect to $\Xi_2 = (\sigma_\zeta, \phi^{\{m\}})$ and $\Xi_1 = (cons, \eta^{\{g\}}, \Theta^{\{k\}}, \alpha^{\{m\}})$. From equation 19, we have:

$$\frac{\partial \mathcal{L}_2}{\partial \Xi_2} = \sum_i \frac{\partial}{\partial \Xi_2} \log \int \prod_{t \in T_i} \prod_{j \in J_t} (s_{ijt}(\zeta; \Xi_2, \hat{\Xi}_1))^{y_{ijt}} dF(\zeta) \quad (34)$$

$$\frac{\partial \mathcal{L}_2}{\partial \Xi_1} = \sum_i \frac{\partial}{\partial \Xi_1} \log \int \prod_{t \in T_i} \prod_{j \in J_t} (s_{ijt}(\zeta; \Xi_2, \hat{\Xi}_1))^{y_{ijt}} dF(\zeta) \quad (35)$$

Note that we can easily calculate the gradients for the outside good choice probabilities because $s_{i0t} = 1 - \sum_l s_{ilt}$. Therefore, the respective gradients are

$$\frac{\partial s_{i0t}}{\partial \gamma} = - \sum_l \frac{\partial s_{ilt}}{\partial \gamma} \quad (36)$$

where $\gamma \in \{\sigma_\zeta, \phi^{\{m\}}, cons, \eta^{\{g\}}, \Theta^{\{k\}}, \alpha^{\{m\}}\}$.

H Technical Details on Counterfactual Simulations

We consider four counterfactual scenarios. Table 12 presents each of the scenarios. When we exclude physicians, we focus on the counterfactual choice probabilities of the affected patients, holding the choices of non-affected patients fixed. We need s_{ijt} and ρ_{ijt} of the affected patients to examine the counterfactual physician choice and opioid prescriptions.

	Did s_{ijt} change?	Did ρ_{ijt} change?	Affected Patients
Exclude and resort	Yes	No	Patients that chose excluded physicians
Exclude and randomize	Yes	No	Patients that chose excluded physicians
Exclude, resort and break bad sorting	Yes, through $\tilde{\rho}$	Yes	Patients that chose excluded physicians
Break bad sorting	Yes, through $\tilde{\rho}$	Yes	Patients that chose inside physicians

Table 12: Counterfactual simulations

H.1 Exclude top % physician and affected patients resort

We simulate the exclusion policy by dropping physicians in the top n^{th} percentile of risk-adjusted prescription rate. We set the mean utility δ_{jt} of the excluded physicians as $-\infty$. We use Gaussian quadrature to calculate the choice probabilities s_{ijt} . Thus, the patient's perceived prescription probabilities $\tilde{\rho}_{ijtm}$ at each integration node m is:

$$\tilde{\rho}_{ijtm} = \frac{\exp(\eta_j + \Theta_i + \tau_t) * \exp(\zeta_{jm})}{1 + \exp(\eta_j + \Theta_i + \tau_t) * \exp(\zeta_{jm})}$$

where ζ_{jm} is the integration node. The patient's choice probabilities s_{ijtm} are:

$$s_{ijtm} = \frac{\exp(\delta_{jt} + \tilde{\rho}_{ijtm} \times (\Theta_i + \tau_t) + Z_{it}\phi)}{1 + \sum_k \exp(\delta_{kt} + \tilde{\rho}_{iktm} \times (\Theta_i + \tau_t) + Z_{it}\phi)}$$

H.2 Exclude top % physician and randomize affected patient choice

For this counterfactual, we use Monte Carlo integration to compute the choice probabilities s_{ijt} . At each Monte Carlo draw m , we have $J + 1$ T1EV errors and one standard normal draw for each patient i , scaled by the dispersion of information type, σ_ζ . The perceived prescription probabilities are:

$$\tilde{\rho}_{ijtm} = \frac{\exp(\eta_j + \Theta_i + \tau_t) * \exp(\zeta_{jm})}{1 + \exp(\eta_j + \Theta_i + \tau_t) * \exp(\zeta_{jm})}$$

The indirect utility of a non-excluded physician is given by:

$$u_{ijtm} = \delta_{jt} + [\tilde{\rho}_{ijtm} * (\Theta_i + \tau_t)] + Z_{it}\phi + \nu_{jtm}$$

We calculate the weighted-average utility of all non-excluded physicians and compare it with the realized T1EV error of the outside option. The weights are the conditional market share of the non-excluded physicians. Patients choose the outside option if and only if:

$$\nu_{0tm} \geq \bar{u}_{itm}$$

where ν_{0tm} is the T1EV of the outside option in market t at MC draw m and $\bar{u}_{itm}$ is the weighted average utility of the non-excluded physicians and takes the form:

$$\bar{u}_{itm} = \sum_{j \notin \text{excluded}} s_{jt}^{cond} \times u_{ijtm}$$

s_{jt}^{cond} is the conditional market share of the non-excluded physicians. We implement this 100 times per patient and use the average across the 100 trials to calculate the counterfactual probability of choosing the outside option, s_{i0t} . We calculate the counterfactual choice probabilities s_{ijt} by multiplying $1 - s_{i0t}$ with the conditional market shares of the non-excluded physicians.

H.3 Exclude top % physician, affected patients resort and break bad sorting

This counterfactual exercise is similar to section H.1, where we exclude physicians and allow patients to resort. Additionally, we require physicians to prescribe only when considering preferences that are concordant with CDC guidelines or medical consensus. Specifically, for each patient affected by the exclusion policy, we require physicians to prescribe based on the following rules:

1. If the patient suffers from sickle cell disease, cancer-related pain, or is in end-of-life care or palliative care, the prescription rates are unchanged:

$$\rho_{ijt} = \frac{\exp \left(\eta_j + \overbrace{X_{i1}\beta_1 + X_{i2}\beta_2 + \theta_i}^{\Theta_i} + \tau_t \right)}{1 + \exp \left(\eta_j + \underbrace{X_{i1}\beta_1 + X_{i2}\beta_2 + \theta_i}_{\Theta_i} + \tau_t \right)}$$

2. In all other cases, if a patient's preference related to bad sorting is positive, that is, the sum of observed preferences concordant with neither the guidelines nor medical consensus, plus a portion of θ_i (depending on assumptions about the share unsupported by medical consensus), the physician should ignore this preference. If the patient's preference related to bad sorting is negative, we do not intervene, and the physician should take this preference into account:

$$\rho_{ijt} = \frac{\exp \left(\eta_j + \overbrace{X_{i1}\beta_1 + \alpha\theta_i}^{\Theta_i} + \tau_t \right)}{1 + \exp \left(\eta_j + \underbrace{X_{i1}\beta_1 + \alpha\theta_i}_{\Theta_i} + \tau_t \right)}$$

where $\alpha \in [0, 1]$ and is the portion of θ_i that is supported by medical consensus. As a result, the choice probabilities s_{ijt} for the second group of patients are:

$$s_{ijt} = \frac{\exp \left(\delta_{jt} + \underbrace{\hat{\rho}_{ijt}}_{f(\hat{\Theta}_i=X_{i1}\hat{\beta}_1+\alpha\theta_i)} \times \left(\underbrace{\Theta_i}_{unchanged} + \tau_t \right) + Z_{it}\phi \right)}{1 + \sum_k \exp \left(\delta_{kt} + \underbrace{\hat{\rho}_{ikt}}_{f(\hat{\Theta}_i=X_{i1}\hat{\beta}_1+\alpha\theta_i)} \times \left(\underbrace{\Theta_i}_{unchanged} + \tau_t \right) + Z_{it}\phi \right)}$$

H.4 Break bad sorting

Similar to section H.3, we require physicians to ignore patient opioid preferences that are neither concordant with CDC guidelines nor with medical consensus if the patient does not suffer from sickle cell disease, cancer-related pain, or end-of-life care or palliative care. We do not exclude any physicians and allow patients to resort, knowing that physicians are now strictly following the CDC guidelines for opioid prescriptions. We examine the counterfactual choice probabilities of patients who originally chose a physician, holding the choices of patients who chose the outside option fixed.

H.5 Decomposing the drop in aggregate prescriptions

Table 13: Excluding top 10% of physicians

	Resorting	No Resorting
$\Delta_{baseline}$ (pp)	-10.8	-11.3
– Forego Care	-2.8 (26%)	-10.4 (92%)
– Switch Physicians	-8.0 (74%)	-1.0 (8%)

Note: To quantify foregone care and switch-physician channels, we implement versions of CF1 and CF2 where we restrict patients to choose the inside options. Denote Y_{all}^k as the outcome of counterfactual k when patients can choose both inside and outside options and Y_{in}^k as the outcome of counterfactual k when patients can only choose the inside options. In the “resorting” column, the switch-physician channel is identified by Y_{in}^1 , and the forego care channel is identified by $Y_{in}^1 - Y_{all}^1$. In the “no resorting” column, the switch physician channel is identified by $Y_{in}^1 - Y_{in}^2$ and the forego care channel is identified by $(Y_{in}^1 - Y_{all}^1) - (Y_{in}^2 - Y_{all}^2)$.